

Pedro vs. Score Base

Eight noisy scenarios. Equal expert-query budgets.
Every run retained, including failures.

Main finding. Score base has lower mean final angular error and ordinary test regret in all eight scenarios. Among the 38 pairs with two valid estimates, it wins 36 pairs on angular error and 36 on regret. This is an empirical comparison of the complete algorithms, not a proof of convergence or a query-policy-only ablation.

31 / 40

Score base final joint successes

4 / 40

Pedro final joint successes

80 / 80

Runs completed through T=20

What counts as success?

Angular error at most 5 degrees **and** mean held-out normalized regret at most 0.01 (1% of the feasible objective range). These thresholds describe approximate recovery, not exact identification of the true parameter.

Study size	Observation model	Evaluation
8 scenarios x 5 seeds x 2 methods	MIN expert; parameter noise; clean $Y=X$	120 held-out queries per test distribution
T=20; 1,600 recorded steps	4D or 6D; no training epochs	All 20 post-observation estimates evaluated

Common setup: $F(\theta,s,x) = \theta \cdot (s \cdot x)$; 120 fixed unit-norm query candidates; repeats allowed. Pedro uses uniform queries and the cone incenter. Score base uses disagreement queries and the mean of 16 parameter samples.

Reading guide

Pages 2-8: final statistics, full-time curves, success and recovery times, balanced tests, failures and runtime.
Pages 9-16: one results sheet per scenario, including every seed.
Pages 17-36: all 1,600 per-step metric records. The companion data bundle preserves full-precision vectors and raw diagnostics.

Prepared from saved results only; no experiment was rerun. Source: aqaPayam/InverseOptimization, Git commit 193b2e9. Report date: 2026-09-03. PDF values are rounded; the data bundle retains full precision. A displayed rounded zero is not necessarily an exact zero.

Final performance at T=20

Mean +/- sample standard deviation. All numerical averages are conditional on a valid estimate. Counts are shown explicitly; failures are not assigned arbitrary angular errors or zero regret.

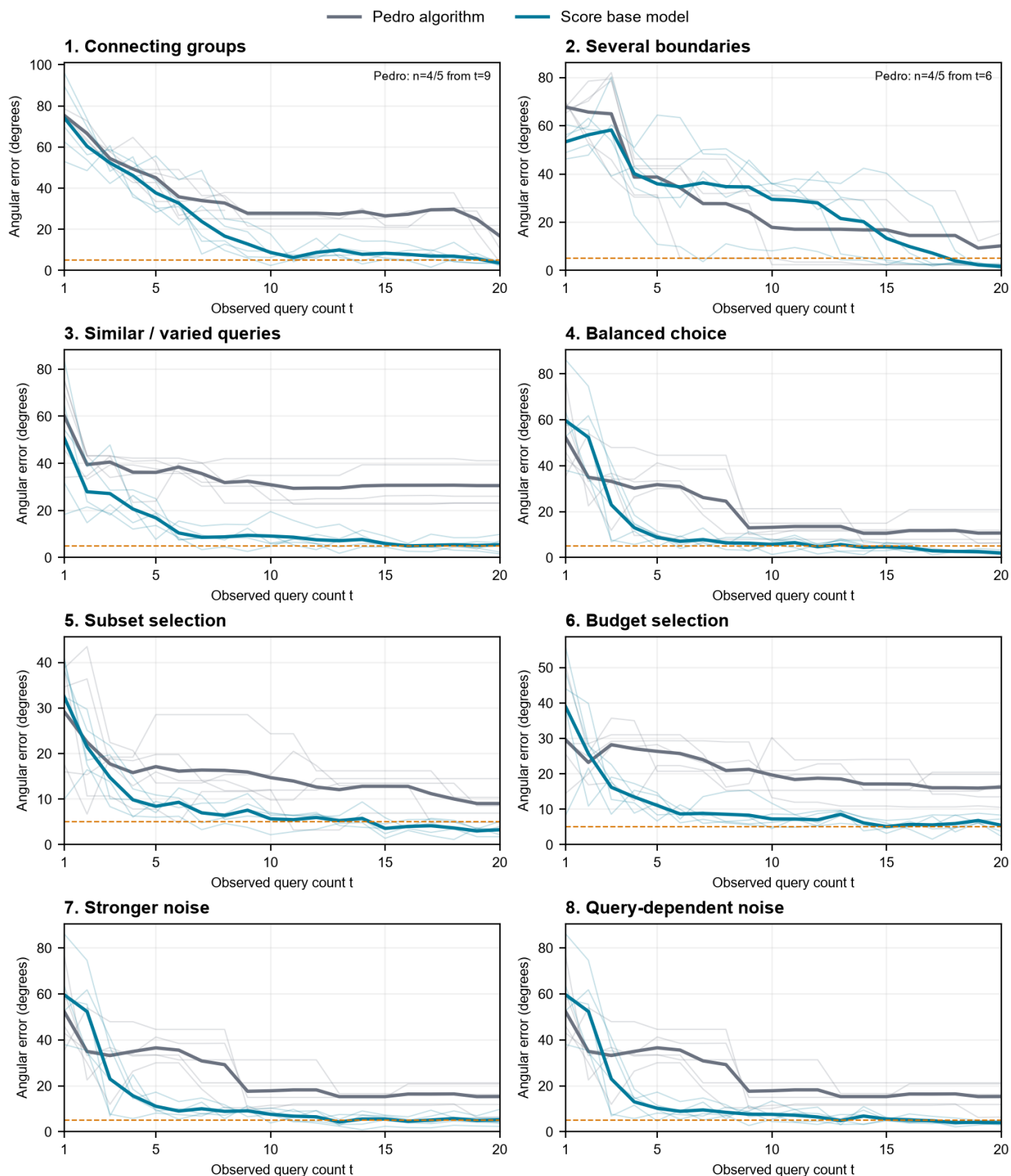
Scenario	Method	Valid	Angle (deg)	Regret (%)	Joint target
1. Connecting groups	Pedro	4/5	16.79 +/- 11.68	1.556 +/- 1.120	1/5
1. Connecting groups	Score	5/5	3.59 +/- 0.93	0.271 +/- 0.155	5/5
2. Several boundaries	Pedro	4/5	10.20 +/- 9.21	2.063 +/- 2.249	2/5
2. Several boundaries	Score	5/5	1.72 +/- 0.85	0.037 +/- 0.050	5/5
3. Similar / varied queries	Pedro	5/5	30.57 +/- 8.96	1.878 +/- 1.353	0/5
3. Similar / varied queries	Score	5/5	5.41 +/- 3.43	0.049 +/- 0.047	2/5
4. Balanced choice	Pedro	5/5	10.67 +/- 6.11	1.007 +/- 0.946	0/5
4. Balanced choice	Score	5/5	2.01 +/- 0.84	0.037 +/- 0.045	5/5
5. Subset selection	Pedro	5/5	9.04 +/- 3.66	0.173 +/- 0.095	1/5
5. Subset selection	Score	5/5	3.27 +/- 1.35	0.032 +/- 0.024	4/5
6. Budget selection	Pedro	5/5	16.25 +/- 4.21	1.048 +/- 0.979	0/5
6. Budget selection	Score	5/5	5.49 +/- 2.45	0.054 +/- 0.024	2/5
7. Stronger noise	Pedro	5/5	15.32 +/- 6.31	1.770 +/- 1.073	0/5
7. Stronger noise	Score	5/5	5.18 +/- 2.91	0.205 +/- 0.142	3/5
8. Query-dependent noise	Pedro	5/5	15.32 +/- 6.31	1.770 +/- 1.073	0/5
8. Query-dependent noise	Score	5/5	3.93 +/- 0.68	0.097 +/- 0.025	5/5

Aggregate outcomes

Method	Ever joint target	Final joint target	Final regret <=1%	Invalid at T
Pedro	6/40	4/40	18/40	2/40
Score base	37/40	31/40	40/40	0/40

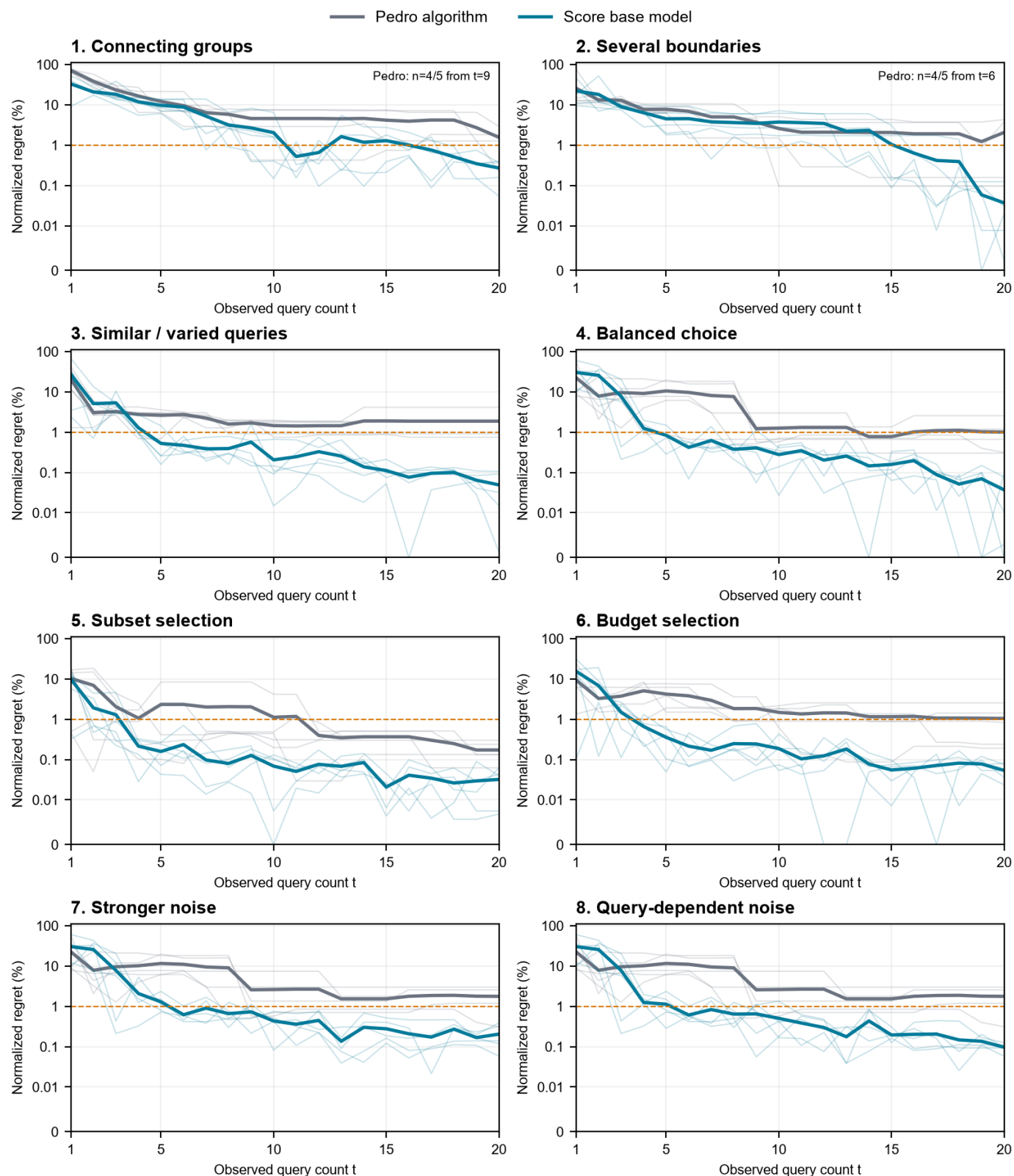
Regret is mean excess true objective cost divided by the feasible cost range at each test query, then averaged. It is not a decision-error percentage and not cumulative training-query regret. Lower is better. 0.01 in the raw data equals 1% in this report.

Angular error over all 20 steps



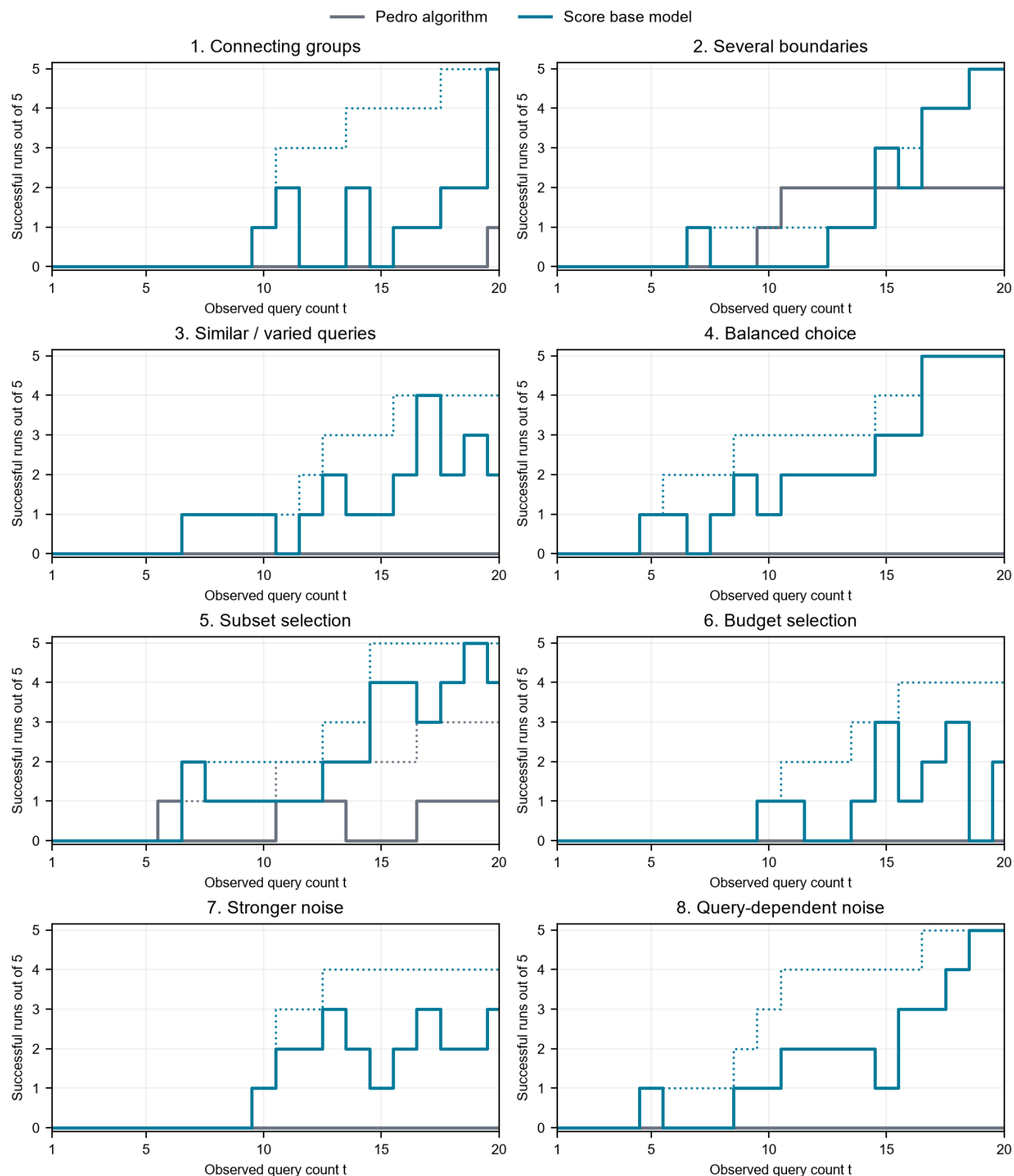
Thin lines are individual seeds; thick lines are valid-only means. Orange dashed line: 5 degrees. Pedro has 4/5 valid runs from t=9 in case 1 and t=6 in case 2; all other plotted counts are 5/5. Curves need not decrease at every step.

Held-out regret over all 20 steps



Ordinary test distributions. Thin lines: seeds; thick lines: valid-only means. Orange dashed line: 1% target. The symlog vertical scale preserves zero while revealing small late-stage differences. Missing invalid estimates are not imputed.

How often is the accuracy target met?



Solid lines: number of runs currently meeting both targets. Dotted lines: number that have ever met both targets by that time. Every count has denominator five; invalid estimates cannot pass. A dotted line staying high does not imply stable recovery.

First versus sustained recovery

Joint target: angle ≤ 5 degrees and ordinary normalized regret ≤ 0.01 . Each list is ordered by seed 0,1,2,3,4. '-' means not reached. All runs continued to T=20; these are retrospective measurements, not actual stopping times.

Scenario	Method	First hit: seeds 0-4	Sustained: seeds 0-4
1. Connecting groups	Pedro	-, -, -, -, 20	-, -, -, -, 20
1. Connecting groups	Score	14, 11, 18, 11, 10	20, 20, 18, 20, 18
2. Several boundaries	Pedro	-, -, 11, 10, -	-, -, 11, 10, -
2. Several boundaries	Score	17, 15, 7, 15, 19	17, 15, 17, 15, 19
3. Similar / varied queries	Pedro	-, -, -, -, -	-, -, -, -, -
3. Similar / varied queries	Score	12, 13, 7, 16, -	-, -, 13, 16, -
4. Balanced choice	Pedro	-, -, -, -, -	-, -, -, -, -
4. Balanced choice	Score	5, 9, 6, 17, 15	17, 15, 11, 17, 15
5. Subset selection	Pedro	17, 6, -, 11, -	17, -, -, -, -
5. Subset selection	Score	13, 7, 15, 15, 7	-, 16, 15, 18, 7
6. Budget selection	Pedro	-, -, -, -, -	-, -, -, -, -
6. Budget selection	Score	-, 14, 11, 16, 10	-, 20, -, -, 20
7. Stronger noise	Pedro	-, -, -, -, -	-, -, -, -, -
7. Stronger noise	Score	13, 10, 11, -, 11	-, 20, 16, -, 11
8. Query-dependent noise	Pedro	-, -, -, -, -	-, -, -, -, -
8. Query-dependent noise	Score	5, 9, 11, 17, 10	18, 19, 16, 17, 14

What this changes in the interpretation

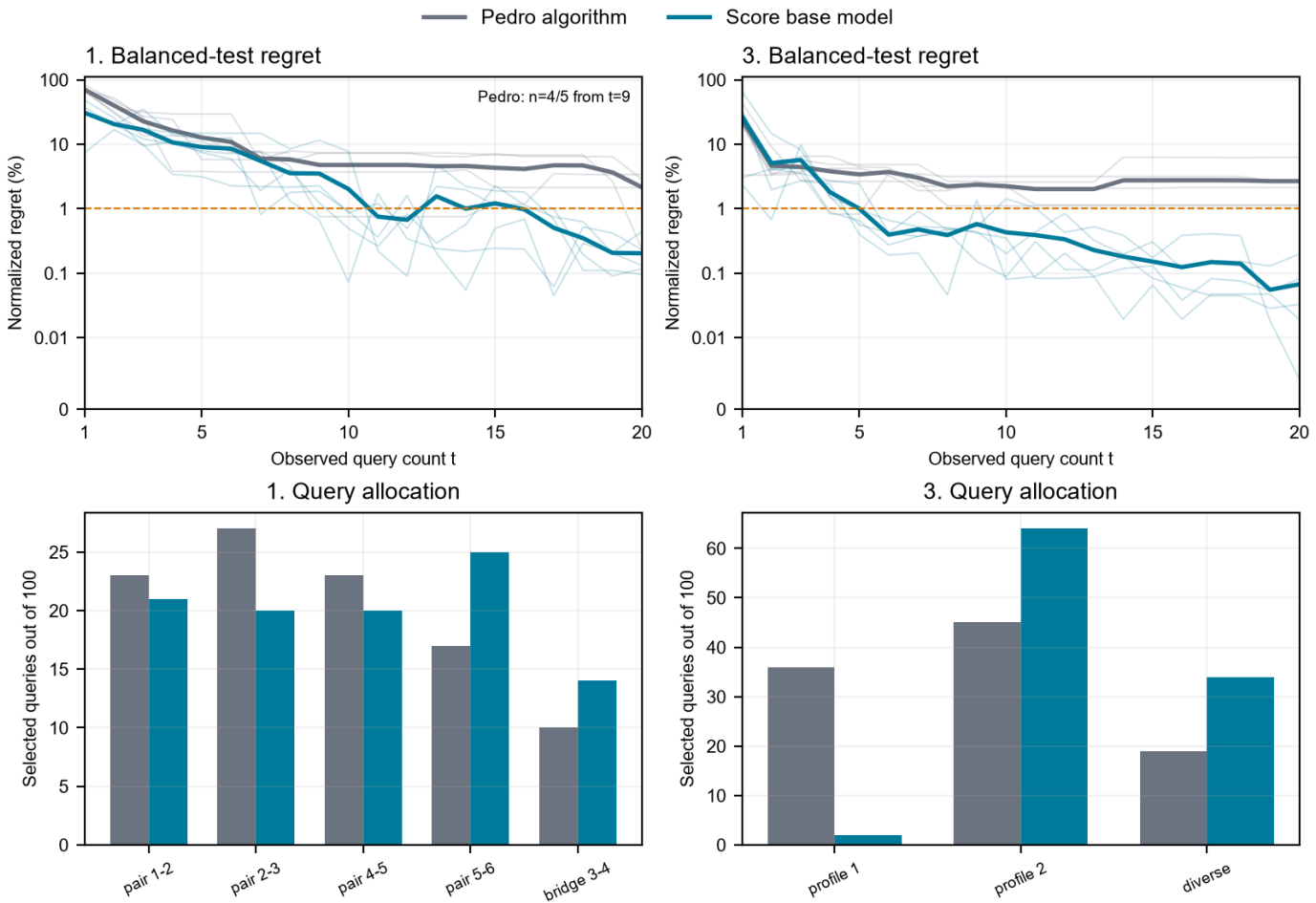
Score base reaches the target at least once in **37/40** runs, but only **31/40** satisfy it at T. Pedro reaches it at least once in **6/40**, with **4/40** passing at T. A single favorable estimate is not the same as stable recovery under noise.

'Sustained from t' means every remaining observed point through T passes. If t=20, only one point has been observed; this does not establish future stability. Comparing medians only among successful runs would also hide the many non-recoveries, so all seed-level times are shown instead.

Example: in case 2, Pedro reaches the target at t=10 for seed 3, before Score base at t=15. Score base is more successful overall, but not uniformly faster on every seed.

Balanced tests and selected queries

The two imbalanced families have additional fresh tests giving equal weight to their predefined groups. These tests are hidden from the algorithms and are not used to choose queries or stop.



Scenario	Pedro regret (%)	Score regret (%)	Valid P; S
1. Connecting groups	2.155 +/- 1.379	0.204 +/- 0.143	4/5; 5/5
3. Similar / varied queries	2.676 +/- 2.116	0.067 +/- 0.079	5/5; 5/5

The final advantage remains on balanced tests. In case 3, Score base selects more diverse queries (34 versus 19 out of 100) and far fewer near profile 1 (2 versus 36). In case 1, connecting-query counts differ more modestly (14 versus 10). The specific query values and the estimator also matter; these counts are not a causal decomposition of the gain.

Reliability, noise and computing cost

Scenario	Pedro seconds	Score seconds	Noise-changed decisions Pedro / Score (each /100)
1. Connecting groups	0.20	13.47	7 / 7
2. Several boundaries	0.07	14.51	5 / 13
3. Similar / varied queries	0.10	14.75	4 / 27
4. Balanced choice	0.08	14.24	0 / 14
5. Subset selection	0.11	14.31	2 / 12
6. Budget selection	0.17	13.16	4 / 7
7. Stronger noise	0.12	18.52	2 / 22
8. Query-dependent noise	0.12	18.42	2 / 23

Mean recorded runtime per full 20-step run is **0.12 s for Pedro** and **15.17 s for Score base**. These timings are the benchmark's run-time measurements, not expert-response costs or a controlled hardware microbenchmark. Score base is better at a fixed query budget here, but computationally slower.

Independent geometric audit of Pedro's failures

Case / seed	First invalid t	Strict interior margin	Nonzero feasible direction
1 / seed 3	9	0 at first failure and T	Yes, both times
2 / seed 4	6	0 at first failure and T	Yes, both times

Independent linear programs in notebook 15 show zero interior margin but remaining nonzero feasible directions. These are **degenerate, lower-dimensional cones**; the returned zero incenter is invalid. The stored legacy message 'no valid nonzero parameter direction' is too strong. Neither a 90/180-degree penalty nor a substitute estimate is reported.

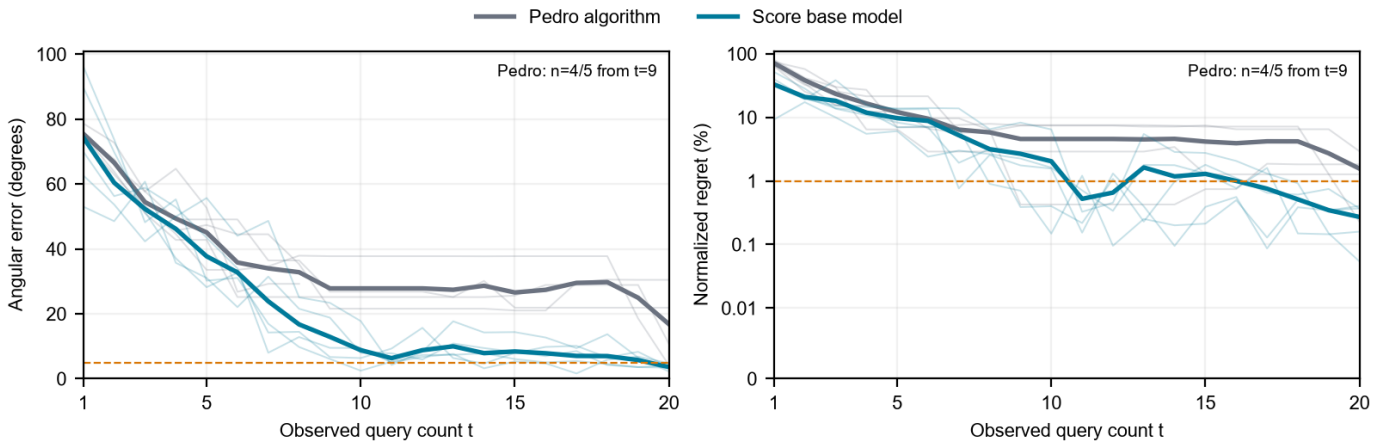
Interpretation limits

Both estimation and query selection differ, so this compares complete algorithms. The sampler uses a finite budget and a loss-based target, not the exact Gaussian-noise likelihood. Five seeds, finite test sets, dimensions 4/6 and this objective family do not establish a general convergence guarantee. Zero test regret does not imply exact parameter recovery.

Noise-changed decisions compare each observed response with the clean MIN response at the selected query. All $Y=X$; none of these differences is observation noise. Active queries may land near decision boundaries and thus be more sensitive to parameter noise.

1. Connecting groups

6D; choose one. Four within-group pairs have 27 candidates each; the connecting pair has 12. Parameter sigma = 0.02.



Seed	Method	Status	Angle (deg)	Regret (%)	Balanced (%)	First joint	Stable joint	Time (s)	Noise flips
0	Pedro	valid	10.73	2.912	2.859	-	-	0.66	3
0	Score	valid	3.59	0.368	0.132	14	20	13.81	1
1	Pedro	valid	30.44	1.819	3.394	-	-	0.09	2
1	Score	valid	2.23	0.054	0.095	11	20	13.73	2
2	Pedro	valid	21.87	1.268	2.128	-	-	0.10	1
2	Score	valid	3.39	0.159	0.117	18	18	13.33	0
3	Pedro	FAIL	-	-	-	-	-	0.05	1
3	Score	valid	4.76	0.404	0.442	11	20	13.14	2
4	Pedro	valid	4.12	0.226	0.239	20	20	0.09	0
4	Score	valid	3.98	0.368	0.232	10	18	13.34	2

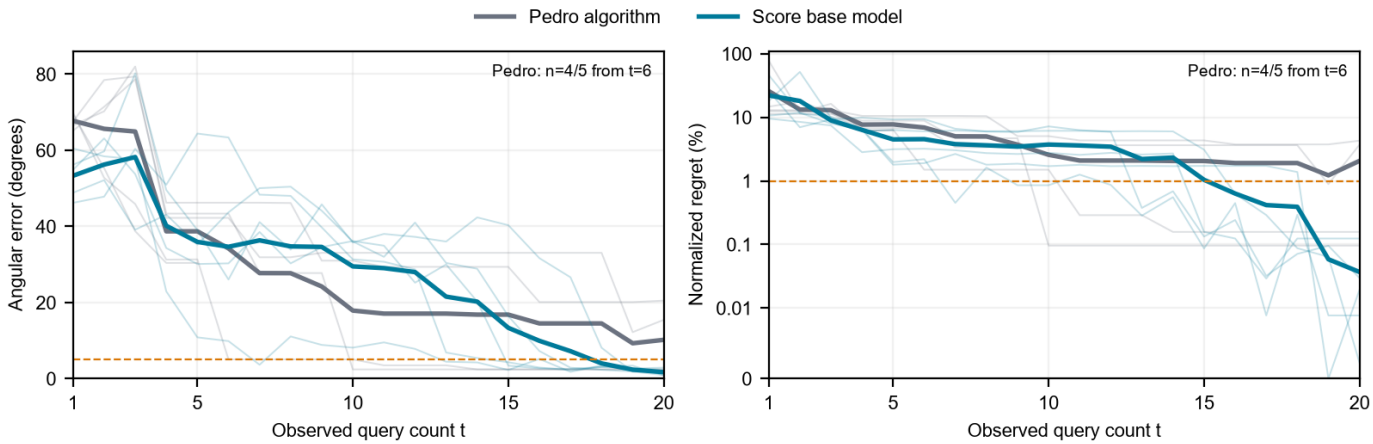
Reading this result

Score base finishes with lower mean error and all five runs meet the joint target. Early target hits are not always stable. Connecting-query counts are 14/100 for Score base and 10/100 for Pedro, so the result is not explained simply by a very large increase in bridge-query frequency.

Each seed has a different hidden parameter, shared by the paired methods. Thin curves are individual seeds; thick curves are valid-only means. Regret percentages use clean held-out decisions. '-' denotes an unavailable metric or an unreached target, not zero. Noise flips are out of 20 responses.

2. Several boundaries

4D; choose one. Three reference-coordinate comparisons have 40 candidate ratios each. Parameter sigma = 0.02.



Seed	Method	Status	Angle (deg)	Regret (%)	Balanced (%)	First joint	Stable joint	Time (s)	Noise flips
0	Pedro	valid	20.50	4.288	-	-	-	0.08	0
0	Score	valid	0.93	0.002	-	17	17	14.48	3
1	Pedro	valid	15.45	3.711	-	-	-	0.08	1
1	Score	valid	0.84	0.030	-	15	15	15.42	2
2	Pedro	valid	2.43	0.157	-	11	11	0.09	0
2	Score	valid	1.88	0.009	-	7	17	13.22	1
3	Pedro	valid	2.43	0.096	-	10	10	0.08	1
3	Score	valid	2.89	0.124	-	15	15	14.54	3
4	Pedro	FAIL	-	-	-	-	-	0.04	3
4	Score	valid	2.03	0.019	-	19	19	14.92	4

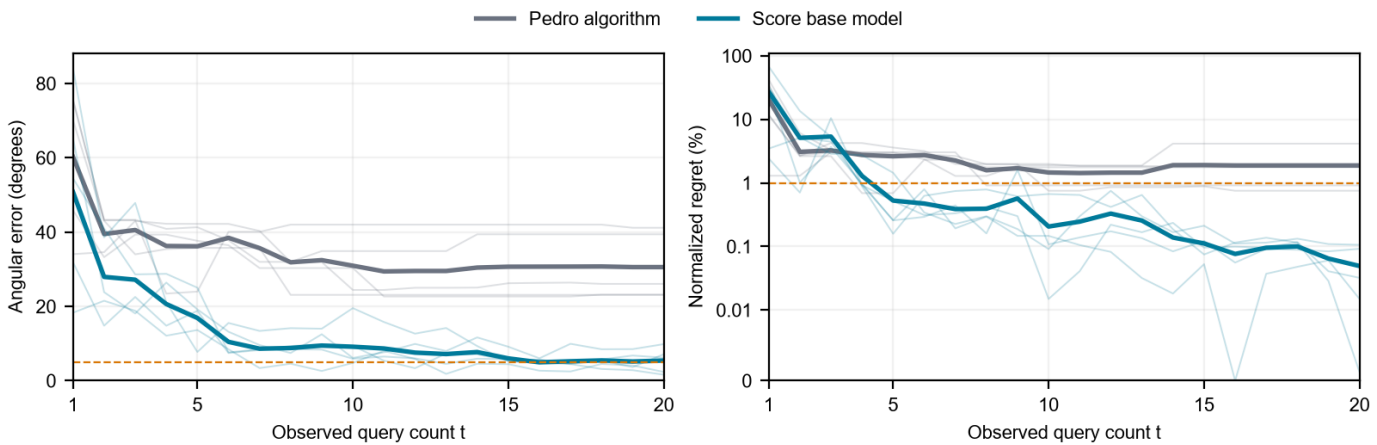
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Pedro is better at t=10 on the valid-only averages (17.89 versus 29.49 degrees), while Score base finishes better (1.72 versus 10.20 degrees). This is evidence against claiming uniform superiority at every intermediate time. One Pedro incenter becomes invalid.

Each seed has a different hidden parameter, shared by the paired methods. Thin curves are individual seeds; thick curves are valid-only means. Regret percentages use clean held-out decisions. '-' denotes an unavailable metric or an unreached target, not zero. Noise flips are out of 20 responses.

3. Similar versus varied queries

6D; choose two. Two fixed query profiles have 48 nearby candidates each; 24 candidates are diverse. Parameter sigma = 0.02.



Seed	Method	Status	Angle (deg)	Regret (%)	Balanced (%)	First joint	Stable joint	Time (s)	Noise flips
0	Pedro	valid	39.39	4.137	6.257	-	-	0.10	1
0	Score	valid	6.18	0.015	0.019	12	-	14.76	5
1	Pedro	valid	23.09	0.912	1.107	-	-	0.12	0
1	Score	valid	7.11	0.106	0.198	13	-	14.89	4
2	Pedro	valid	23.15	1.835	1.149	-	-	0.10	2
2	Score	valid	2.34	0.032	0.033	7	13	15.33	8
3	Pedro	valid	41.15	1.756	2.729	-	-	0.10	0
3	Score	valid	1.60	0.001	0.004	16	16	14.94	5
4	Pedro	valid	26.04	0.751	2.137	-	-	0.10	1
4	Score	valid	9.83	0.092	0.082	-	-	13.85	5

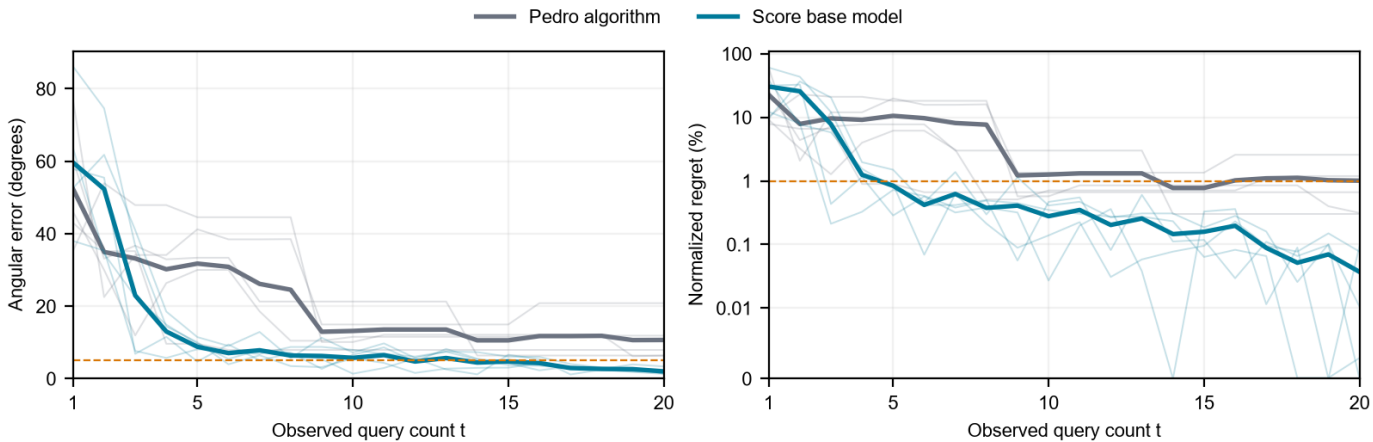
Reading this result

Pedro largely plateaus near 30 degrees, while Score base continues improving. Score base chooses 34/100 diverse queries versus Pedro's 19/100. Only two Score base runs finish within 5 degrees despite all five having low regret: decision quality and parameter recovery are different outcomes.

Each seed has a different hidden parameter, shared by the paired methods. Thin curves are individual seeds; thick curves are valid-only means. Regret percentages use clean held-out decisions. '-' denotes an unavailable metric or an unreached target, not zero. Noise flips are out of 20 responses.

4. Ordinary balanced choice

4D; choose one. Dense unit-sphere query candidates. Parameter sigma = 0.02.



Seed	Method	Status	Angle (deg)	Regret (%)	Balanced (%)	First joint	Stable joint	Time (s)	Noise flips
0	Pedro	valid	7.92	0.663	-	-	-	0.08	0
0	Score	valid	1.31	0.095	-	5	17	14.38	2
1	Pedro	valid	20.84	2.571	-	-	-	0.08	0
1	Score	valid	2.04	0.000	-	9	15	14.21	3
2	Pedro	valid	6.38	0.302	-	-	-	0.09	0
2	Score	valid	3.38	0.077	-	6	11	14.19	3
3	Pedro	valid	11.86	1.189	-	-	-	0.08	0
3	Score	valid	1.36	0.010	-	17	17	13.51	1
4	Pedro	valid	6.36	0.313	-	-	-	0.09	0
4	Score	valid	1.96	0.003	-	15	15	14.90	5

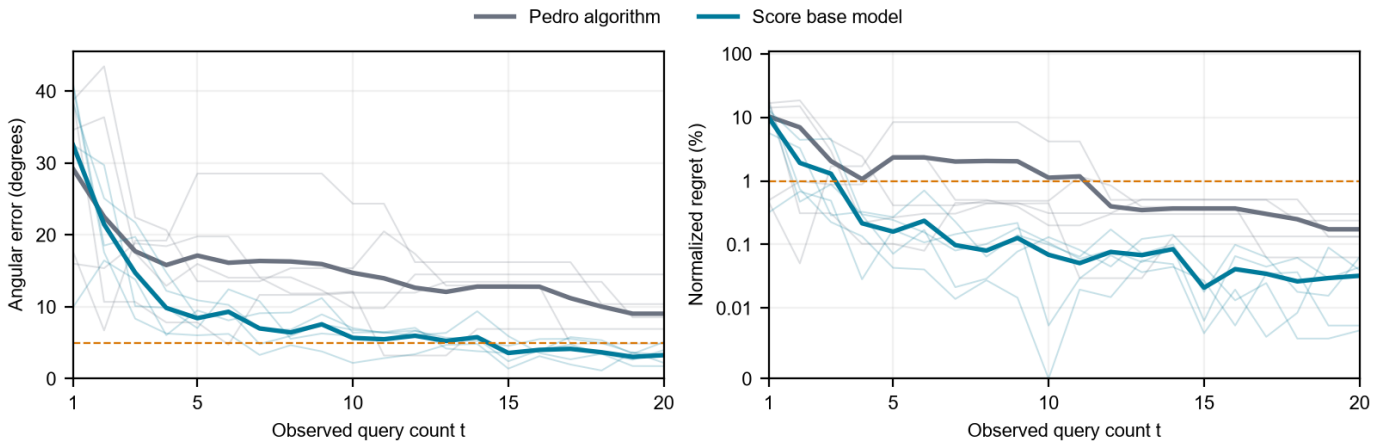
Reading this result

The advantage persists without a deliberately rare query group. Score base finishes near 2 degrees on average; all five runs pass the joint target. Pedro finishes near 10.7 degrees and none pass the angular target. One Score base run has zero final regret on the finite test set, not exact parameter recovery.

Each seed has a different hidden parameter, shared by the paired methods. Thin curves are individual seeds; thick curves are valid-only means. Regret percentages use clean held-out decisions. '-' denotes an unavailable metric or an unreached target, not zero. Noise flips are out of 20 responses.

5. Ordinary subset selection

6D; choose exactly three of six. Dense unit-sphere query candidates. Parameter sigma = 0.02.



Seed	Method	Status	Angle (deg)	Regret (%)	Balanced (%)	First joint	Stable joint	Time (s)	Noise flips
0	Pedro	valid	4.88	0.063	-	17	17	0.11	1
0	Score	valid	5.09	0.008	-	13	-	14.52	3
1	Pedro	valid	8.52	0.132	-	6	-	0.10	0
1	Score	valid	1.74	0.007	-	7	16	13.65	3
2	Pedro	valid	14.50	0.301	-	-	-	0.10	0
2	Score	valid	3.97	0.063	-	15	15	15.40	5
3	Pedro	valid	6.91	0.134	-	11	-	0.11	1
3	Score	valid	3.35	0.044	-	15	18	14.02	0
4	Pedro	valid	10.37	0.237	-	-	-	0.11	0
4	Score	valid	2.20	0.039	-	7	7	13.97	1

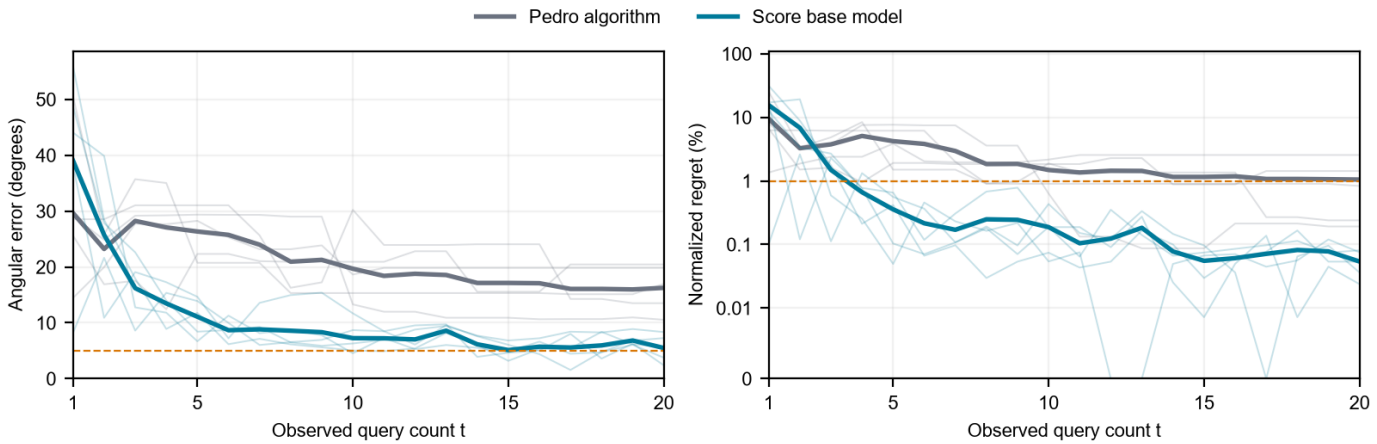
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This is the strongest ordinary case for Pedro's decision quality: both methods finish below 1% regret in every run. Score base still estimates the parameter more accurately on average. Pedro has slightly lower final angular error for seed 0, but Score base has lower regret for that seed.

Each seed has a different hidden parameter, shared by the paired methods. Thin curves are individual seeds; thick curves are valid-only means. Regret percentages use clean held-out decisions. '-' denotes an unavailable metric or an unreached target, not zero. Noise flips are out of 20 responses.

6. Budget-constrained selection

6D; binary knapsack. Weights [1,2,2,3,3,4], capacity 6; exact small-set MIN. Parameter sigma = 0.02.



Seed	Method	Status	Angle (deg)	Regret (%)	Balanced (%)	First joint	Stable joint	Time (s)	Noise flips
0	Pedro	valid	20.44	1.429	-	-	-	0.16	0
0	Score	valid	8.34	0.081	-	-	-	13.42	4
1	Pedro	valid	13.51	0.239	-	-	-	0.17	2
1	Score	valid	2.39	0.024	-	14	20	13.98	1
2	Pedro	valid	19.83	2.551	-	-	-	0.17	0
2	Score	valid	5.71	0.054	-	11	-	12.89	1
3	Pedro	valid	10.54	0.192	-	-	-	0.17	0
3	Score	valid	7.26	0.036	-	16	-	12.64	1
4	Pedro	valid	16.96	0.831	-	-	-	0.17	2
4	Score	valid	3.75	0.075	-	10	20	12.85	0

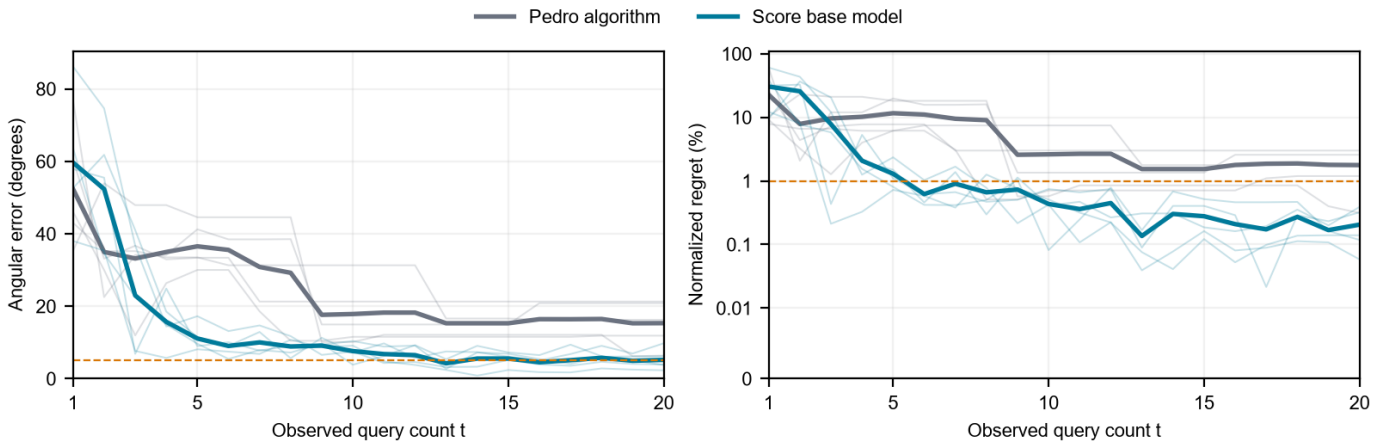
Reading this result

Score base substantially improves final regret, but only two of five runs finish within 5 degrees. All five nevertheless pass the regret target. The small knapsack MIN is solved exactly; no large training job or approximate forward surrogate is involved.

Each seed has a different hidden parameter, shared by the paired methods. Thin curves are individual seeds; thick curves are valid-only means. Regret percentages use clean held-out decisions. '-' denotes an unavailable metric or an unreached target, not zero. Noise flips are out of 20 responses.

7. Stronger parameter noise

4D; same true parameters, candidates and tests as case 4. Parameter sigma increases to 0.08.



Seed	Method	Status	Angle (deg)	Regret (%)	Balanced (%)	First joint	Stable joint	Time (s)	Noise flips
0	Pedro	valid	16.25	1.781	-	-	-	0.08	1
0	Score	valid	6.18	0.320	-	13	-	15.34	4
1	Pedro	valid	20.84	2.571	-	-	-	0.17	0
1	Score	valid	3.77	0.118	-	10	20	22.19	3
2	Pedro	valid	21.28	2.996	-	-	-	0.14	1
2	Score	valid	3.90	0.058	-	11	16	20.83	3
3	Pedro	valid	11.86	1.189	-	-	-	0.09	0
3	Score	valid	9.77	0.389	-	-	-	14.82	5
4	Pedro	valid	6.36	0.313	-	-	-	0.10	0
4	Score	valid	2.31	0.140	-	11	11	19.43	7

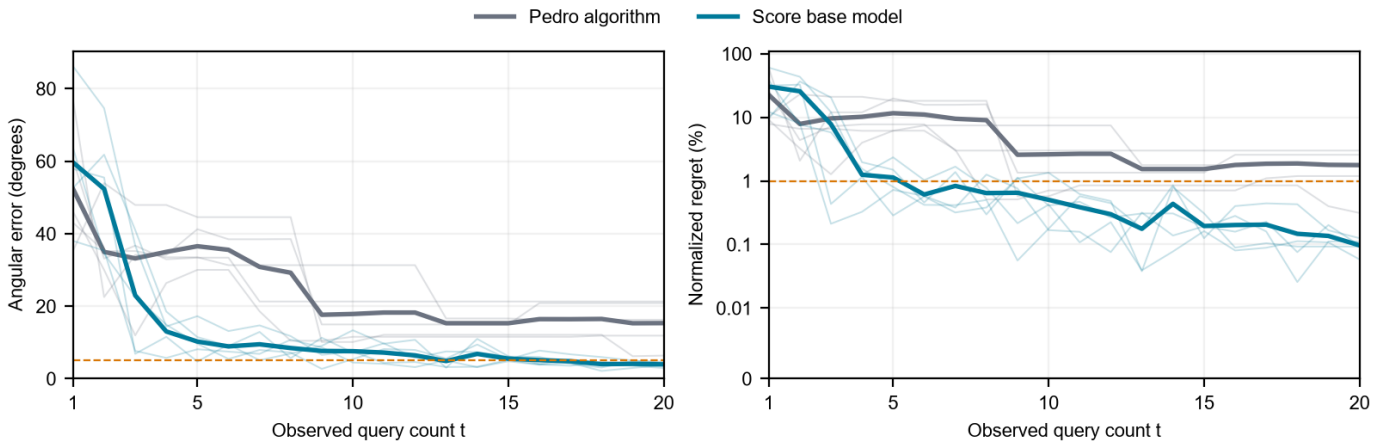
Reading this result

Compared with case 4, stronger parameter noise raises Score base's mean final angular error from 2.01 to 5.18 degrees. It still outperforms Pedro, but only three of five runs meet the joint target at T. Active queries can be informative and also sensitive to parameter perturbations.

Each seed has a different hidden parameter, shared by the paired methods. Thin curves are individual seeds; thick curves are valid-only means. Regret percentages use clean held-out decisions. '-' denotes an unavailable metric or an unreached target, not zero. Noise flips are out of 20 responses.

8. Query-dependent noise

4D; same true parameters, candidates and tests as case 4. Parameter $\sigma(s) = 0.02 + 0.08 \cdot \text{abs}(s[0])$.



Seed	Method	Status	Angle (deg)	Regret (%)	Balanced (%)	First joint	Stable joint	Time (s)	Noise flips
0	Pedro	valid	16.25	1.781	-	-	-	0.15	1
0	Score	valid	2.93	0.093	-	5	18	14.16	5
1	Pedro	valid	20.84	2.571	-	-	-	0.13	0
1	Score	valid	4.38	0.125	-	9	19	20.29	7
2	Pedro	valid	21.28	2.996	-	-	-	0.08	1
2	Score	valid	3.90	0.058	-	11	16	19.55	3
3	Pedro	valid	11.86	1.189	-	-	-	0.09	0
3	Score	valid	4.71	0.111	-	17	17	18.21	2
4	Pedro	valid	6.36	0.313	-	-	-	0.13	0
4	Score	valid	3.75	0.096	-	10	14	19.87	6

Reading this result

All five Score base runs meet the final joint target. Pedro's full query/observation sequences match case 7 for these seeds, despite different perturbed expert parameters; its identical results are therefore expected. Different parameter perturbations need not change a discrete MIN decision.

Each seed has a different hidden parameter, shared by the paired methods. Thin curves are individual seeds; thick curves are valid-only means. Regret percentages use clean held-out decisions. '-' denotes an unavailable metric or an unreached target, not zero. Noise flips are out of 20 responses.

Complete step records | runs 1-4 of 80

1. Connecting groups | seed 0 | Pedro

t	Angle (deg)	Regret (%)	Balanced (%)	Status
1	78.619	65.4638	68.9373	valid
2	72.731	32.3927	37.9002	valid
3	57.689	25.5846	31.3352	valid
4	49.057	21.4213	29.4452	valid
5	49.057	21.4213	29.4452	valid
6	49.057	21.4213	29.4452	valid
7	36.428	6.1081	5.4654	valid
8	36.428	6.1081	5.4654	valid
9	26.815	7.4137	7.3071	valid
10	26.815	7.4137	7.3071	valid
11	26.815	7.4137	7.3071	valid
12	26.815	7.4137	7.3071	valid
13	25.129	7.1379	6.4330	valid
14	25.129	7.1379	6.4330	valid
15	25.596	7.4482	7.1463	valid
16	28.862	6.4861	6.4426	valid
17	28.862	6.4861	6.4426	valid
18	28.862	6.4861	6.4426	valid
19	28.862	6.4861	6.4426	valid
20	10.731	2.9122	2.8589	valid

1. Connecting groups | seed 1 | Pedro

t	Angle (deg)	Regret (%)	Balanced (%)	Status
1	73.280	64.0993	66.5905	valid
2	62.911	25.3285	31.8533	valid
3	52.438	16.1648	15.5858	valid
4	45.636	12.3210	10.9971	valid
5	47.339	12.3210	10.9971	valid
6	44.482	9.6381	7.7112	valid
7	44.482	9.6381	7.7112	valid
8	35.295	6.3015	4.6207	valid
9	21.559	0.4252	0.7536	valid
10	21.559	0.4252	0.7536	valid
11	21.559	0.4252	0.7536	valid
12	21.559	0.4252	0.7536	valid
13	21.559	0.4252	0.7536	valid
14	21.559	0.4252	0.7536	valid
15	21.005	0.7412	1.1802	valid
16	21.005	0.7412	1.1802	valid
17	29.584	1.8573	3.4970	valid
18	30.520	1.8186	3.3941	valid
19	30.442	1.8186	3.3941	valid
20	30.442	1.8186	3.3941	valid

1. Connecting groups | seed 0 | Score base

t	Angle (deg)	Regret (%)	Balanced (%)	Status
1	89.733	51.7053	48.2243	valid
2	69.796	28.3148	25.5858	valid
3	51.357	13.6197	9.4482	valid
4	37.091	10.8320	10.9570	valid
5	28.189	8.2867	7.3821	valid
6	32.626	6.9498	5.8829	valid
7	14.215	1.9252	1.9258	valid
8	14.436	3.1792	4.0341	valid
9	6.665	0.3872	1.2247	valid
10	6.340	0.3996	0.4970	valid
11	9.272	0.2171	0.2606	valid
12	15.654	1.0741	0.8341	valid
13	6.345	0.2288	0.2920	valid
14	4.638	0.9720	0.5664	valid
15	7.930	1.7902	2.2884	valid
16	9.773	1.1023	1.1375	valid
17	10.109	0.6491	0.7482	valid
18	6.529	0.6081	0.6271	valid
19	8.275	0.7463	0.2039	valid
20	3.585	0.3676	0.1316	valid

1. Connecting groups | seed 1 | Score base

t	Angle (deg)	Regret (%)	Balanced (%)	Status
1	62.604	30.9176	29.4303	valid
2	53.644	19.4651	20.7904	valid
3	42.295	15.2398	17.3396	valid
4	49.927	14.5605	14.9679	valid
5	55.654	13.7866	14.8132	valid
6	44.007	13.9740	14.8132	valid
7	48.612	13.7866	14.8132	valid
8	25.043	6.4980	8.4701	valid
9	23.361	8.2493	11.5091	valid
10	17.742	6.4402	7.7524	valid
11	4.227	0.3263	0.2208	valid
12	8.290	0.4534	0.0900	valid
13	17.669	5.4609	4.8369	valid
14	14.230	2.8453	2.2067	valid
15	14.343	2.7476	1.8935	valid
16	12.678	2.0558	1.8210	valid
17	9.404	1.3038	0.6776	valid
18	13.691	0.9145	0.1103	valid
19	6.490	0.1497	0.1103	valid
20	2.226	0.0539	0.0955	valid

Complete step records | runs 5-8 of 80

1. Connecting groups | seed 2 | Pedro

t	Angle (deg)	Regret (%)	Balanced (%)	Status
1	75.061	74.3193	67.7776	valid
2	63.782	57.4038	52.1548	valid
3	50.956	29.7385	26.0332	valid
4	42.765	6.4416	3.7707	valid
5	42.765	6.4416	3.7707	valid
6	25.149	2.8901	3.6973	valid
7	25.149	2.8901	3.6973	valid
8	25.149	2.8901	3.6973	valid
9	25.149	2.8901	3.6973	valid
10	25.149	2.8901	3.6973	valid
11	25.149	2.8901	3.6973	valid
12	25.149	2.8901	3.6973	valid
13	25.149	2.8901	3.6973	valid
14	30.116	3.4037	4.0471	valid
15	21.867	1.2682	2.1279	valid
16	21.867	1.2682	2.1279	valid
17	21.867	1.2682	2.1279	valid
18	21.867	1.2682	2.1279	valid
19	21.867	1.2682	2.1279	valid
20	21.867	1.2682	2.1279	valid

1. Connecting groups | seed 3 | Pedro

t	Angle (deg)	Regret (%)	Balanced (%)	Status
1	75.467	72.0266	65.9488	valid
2	65.839	33.9157	30.6760	valid
3	57.663	15.3338	13.5339	valid
4	64.683	15.3338	13.5339	valid
5	52.626	13.2343	13.5276	valid
6	26.748	6.6231	7.5521	valid
7	29.218	5.7542	7.3925	valid
8	29.218	5.7542	7.3925	valid
9	-	-	-	FAIL
10	-	-	-	FAIL
11	-	-	-	FAIL
12	-	-	-	FAIL
13	-	-	-	FAIL
14	-	-	-	FAIL
15	-	-	-	FAIL
16	-	-	-	FAIL
17	-	-	-	FAIL
18	-	-	-	FAIL
19	-	-	-	FAIL
20	-	-	-	FAIL

1. Connecting groups | seed 2 | Score base

t	Angle (deg)	Regret (%)	Balanced (%)	Status
1	52.991	9.2071	7.3840	valid
2	48.524	17.2997	16.7372	valid
3	60.702	9.9686	9.9051	valid
4	35.692	5.4933	3.4226	valid
5	31.026	6.0554	3.1293	valid
6	22.046	2.3990	2.2663	valid
7	31.436	2.9911	2.2105	valid
8	21.535	2.6355	2.1490	valid
9	18.800	2.2439	2.2728	valid
10	7.929	1.6152	0.8618	valid
11	7.801	0.7077	1.2027	valid
12	5.903	0.3377	0.4969	valid
13	10.846	1.7845	2.2160	valid
14	9.425	1.7655	1.9592	valid
15	8.262	1.3029	1.1322	valid
16	6.935	0.7685	0.9945	valid
17	8.493	1.5979	0.9945	valid
18	4.209	0.1490	0.1934	valid
19	3.542	0.1446	0.0905	valid
20	3.389	0.1589	0.1172	valid

1. Connecting groups | seed 3 | Score base

t	Angle (deg)	Regret (%)	Balanced (%)	Status
1	69.939	33.3343	31.5337	valid
2	56.249	19.1047	19.5504	valid
3	58.813	38.3153	34.8402	valid
4	52.831	15.8146	14.0008	valid
5	43.423	13.3449	11.9460	valid
6	33.975	13.4295	12.6136	valid
7	7.985	0.7604	0.8199	valid
8	12.828	2.5528	1.7863	valid
9	9.494	1.7520	1.8705	valid
10	9.431	1.5600	0.9142	valid
11	4.983	0.1533	0.3628	valid
12	6.696	1.3072	1.5960	valid
13	7.614	0.3983	0.2031	valid
14	3.189	0.0949	0.0549	valid
15	5.305	0.3894	0.5020	valid
16	4.721	0.5577	0.6863	valid
17	1.603	0.0865	0.0448	valid
18	5.838	0.3845	0.2910	valid
19	6.782	0.3427	0.2079	valid
20	4.762	0.4039	0.4419	valid

Complete step records | runs 9-12 of 80

1. Connecting groups | seed 4 | Pedro

t	Angle (deg)	Regret (%)	Balanced (%)	Status
1	74.839	80.2591	82.5148	valid
2	67.629	41.6503	47.6794	valid
3	53.311	30.0070	27.0106	valid
4	44.582	27.0374	24.1069	valid
5	33.525	6.9562	5.7735	valid
6	33.525	6.9562	5.7735	valid
7	34.604	7.5693	5.7941	valid
8	37.894	7.8996	7.9401	valid
9	37.735	7.5160	7.3269	valid
10	37.735	7.5160	7.3269	valid
11	37.735	7.5160	7.3269	valid
12	37.735	7.5160	7.3269	valid
13	37.735	7.5160	7.3269	valid
14	37.706	7.3572	7.1672	valid
15	37.739	7.1420	6.7876	valid
16	37.739	7.1420	6.7876	valid
17	37.739	7.1420	6.7876	valid
18	37.739	7.1420	6.7876	valid
19	18.549	1.2768	2.6737	valid
20	4.125	0.2257	0.2387	valid

2. Several boundaries | seed 0 | Pedro

t	Angle (deg)	Regret (%)	Balanced (%)	Status
1	66.583	12.9290	-	valid
2	70.236	12.9290	-	valid
3	82.003	12.9290	-	valid
4	46.228	10.5273	-	valid
5	46.228	10.5273	-	valid
6	46.228	10.5273	-	valid
7	46.228	10.5273	-	valid
8	46.142	10.4337	-	valid
9	31.000	5.0525	-	valid
10	31.000	5.0525	-	valid
11	29.373	4.3297	-	valid
12	29.373	4.3297	-	valid
13	29.373	4.3297	-	valid
14	29.373	4.3297	-	valid
15	29.373	4.3297	-	valid
16	20.111	3.7433	-	valid
17	20.111	3.7433	-	valid
18	20.111	3.7433	-	valid
19	20.111	3.7433	-	valid
20	20.498	4.2876	-	valid

1. Connecting groups | seed 4 | Score base

t	Angle (deg)	Regret (%)	Balanced (%)	Status
1	96.201	39.8538	37.3276	valid
2	73.696	20.3930	20.1039	valid
3	48.179	13.8954	12.1242	valid
4	55.246	12.4837	10.0557	valid
5	30.192	6.9631	7.8098	valid
6	31.046	7.1251	7.1511	valid
7	16.930	6.6084	7.8207	valid
8	9.754	0.8969	1.3224	valid
9	6.074	0.6995	0.6695	valid
10	2.451	0.1481	0.0733	valid
11	5.286	1.2038	1.7161	valid
12	7.311	0.0953	0.3406	valid
13	7.329	0.2517	0.2399	valid
14	7.846	0.1998	0.2175	valid
15	6.038	0.2123	0.2441	valid
16	5.024	0.4972	0.2361	valid
17	5.434	0.1284	0.0617	valid
18	4.505	0.5033	0.5214	valid
19	3.558	0.3446	0.4193	valid
20	3.979	0.3685	0.2319	valid

2. Several boundaries | seed 0 | Score base

t	Angle (deg)	Regret (%)	Balanced (%)	Status
1	48.834	18.0875	-	valid
2	52.150	51.6848	-	valid
3	39.125	9.7973	-	valid
4	43.105	6.0977	-	valid
5	35.224	6.2872	-	valid
6	43.684	6.0070	-	valid
7	50.043	6.1728	-	valid
8	50.434	6.0141	-	valid
9	44.117	6.0523	-	valid
10	36.037	6.1275	-	valid
11	32.002	6.1275	-	valid
12	41.007	6.0070	-	valid
13	30.397	6.0524	-	valid
14	28.873	5.9956	-	valid
15	16.438	3.0967	-	valid
16	7.257	0.6258	-	valid
17	2.870	0.2903	-	valid
18	2.338	0.0873	-	valid
19	1.868	0.0637	-	valid
20	0.935	0.0022	-	valid

Complete step records | runs 13-16 of 80

2. Several boundaries | seed 1 | Pedro

t	Angle (deg)	Regret (%)	Balanced (%)	Status
1	67.561	12.4799	-	valid
2	78.455	12.4799	-	valid
3	79.392	12.4799	-	valid
4	42.239	6.7285	-	valid
5	42.239	6.7285	-	valid
6	42.239	6.7285	-	valid
7	31.896	3.3804	-	valid
8	31.896	3.3804	-	valid
9	33.061	3.6317	-	valid
10	33.061	3.6317	-	valid
11	33.061	3.6317	-	valid
12	33.061	3.6317	-	valid
13	33.061	3.6317	-	valid
14	33.061	3.6317	-	valid
15	33.061	3.6317	-	valid
16	33.061	3.6317	-	valid
17	33.061	3.6317	-	valid
18	33.061	3.6317	-	valid
19	12.218	0.8894	-	valid
20	15.451	3.7115	-	valid

2. Several boundaries | seed 2 | Pedro

t	Angle (deg)	Regret (%)	Balanced (%)	Status
1	69.966	14.6968	-	valid
2	52.771	17.5462	-	valid
3	45.878	11.8249	-	valid
4	31.280	6.0878	-	valid
5	31.332	6.3829	-	valid
6	5.058	1.4967	-	valid
7	5.058	1.4967	-	valid
8	5.058	1.4967	-	valid
9	5.058	1.4967	-	valid
10	5.058	1.4967	-	valid
11	3.495	0.2888	-	valid
12	3.495	0.2888	-	valid
13	3.495	0.2888	-	valid
14	2.431	0.1573	-	valid
15	2.431	0.1573	-	valid
16	2.431	0.1573	-	valid
17	2.431	0.1573	-	valid
18	2.431	0.1573	-	valid
19	2.431	0.1573	-	valid
20	2.431	0.1573	-	valid

2. Several boundaries | seed 1 | Score base

t	Angle (deg)	Regret (%)	Balanced (%)	Status
1	46.195	25.9227	-	valid
2	47.819	6.9702	-	valid
3	60.372	9.2410	-	valid
4	50.939	9.8598	-	valid
5	64.392	9.2410	-	valid
6	63.360	9.3420	-	valid
7	48.346	6.5997	-	valid
8	48.017	5.9616	-	valid
9	39.744	5.8606	-	valid
10	31.351	7.1928	-	valid
11	30.708	6.2717	-	valid
12	28.710	5.8926	-	valid
13	6.861	0.3751	-	valid
14	5.413	0.6862	-	valid
15	4.309	0.1330	-	valid
16	2.916	0.2422	-	valid
17	2.236	0.0320	-	valid
18	3.080	0.0710	-	valid
19	3.413	0.0954	-	valid
20	0.844	0.0296	-	valid

2. Several boundaries | seed 2 | Score base

t	Angle (deg)	Regret (%)	Balanced (%)	Status
1	60.460	46.1005	-	valid
2	58.407	11.4933	-	valid
3	57.644	10.2162	-	valid
4	22.922	6.5202	-	valid
5	10.854	1.9738	-	valid
6	9.865	2.1915	-	valid
7	3.658	0.4510	-	valid
8	11.085	1.6177	-	valid
9	8.865	0.8543	-	valid
10	8.147	0.8543	-	valid
11	9.522	1.2527	-	valid
12	7.819	0.8633	-	valid
13	4.448	0.2839	-	valid
14	4.264	0.5530	-	valid
15	2.283	0.0866	-	valid
16	5.174	0.4466	-	valid
17	1.789	0.0089	-	valid
18	3.458	0.2929	-	valid
19	2.080	0.0089	-	valid
20	1.877	0.0089	-	valid

Complete step records | runs 17-20 of 80

2. Several boundaries | seed 3 | Pedro

t	Angle (deg)	Regret (%)	Balanced (%)	Status
1	65.158	11.2789	-	valid
2	71.420	11.2789	-	valid
3	78.628	11.2789	-	valid
4	43.364	8.8400	-	valid
5	43.364	8.8400	-	valid
6	43.364	8.8400	-	valid
7	27.747	4.6824	-	valid
8	27.747	4.6824	-	valid
9	27.747	4.6824	-	valid
10	2.426	0.0958	-	valid
11	2.426	0.0958	-	valid
12	2.426	0.0958	-	valid
13	2.426	0.0958	-	valid
14	2.426	0.0958	-	valid
15	2.426	0.0958	-	valid
16	2.426	0.0958	-	valid
17	2.426	0.0958	-	valid
18	2.426	0.0958	-	valid
19	2.426	0.0958	-	valid
20	2.426	0.0958	-	valid

2. Several boundaries | seed 4 | Pedro

t	Angle (deg)	Regret (%)	Balanced (%)	Status
1	69.440	77.4829	-	valid
2	55.334	11.9181	-	valid
3	38.744	16.2865	-	valid
4	30.399	6.1882	-	valid
5	30.399	6.1882	-	valid
6	-	-	-	FAIL
7	-	-	-	FAIL
8	-	-	-	FAIL
9	-	-	-	FAIL
10	-	-	-	FAIL
11	-	-	-	FAIL
12	-	-	-	FAIL
13	-	-	-	FAIL
14	-	-	-	FAIL
15	-	-	-	FAIL
16	-	-	-	FAIL
17	-	-	-	FAIL
18	-	-	-	FAIL
19	-	-	-	FAIL
20	-	-	-	FAIL

2. Several boundaries | seed 3 | Score base

t	Angle (deg)	Regret (%)	Balanced (%)	Status
1	54.842	9.5462	-	valid
2	63.113	8.4725	-	valid
3	53.747	7.4359	-	valid
4	34.311	2.8241	-	valid
5	30.121	3.1440	-	valid
6	30.292	3.2036	-	valid
7	38.525	3.0332	-	valid
8	30.287	2.7333	-	valid
9	34.567	2.6450	-	valid
10	36.152	2.7609	-	valid
11	34.938	2.5623	-	valid
12	25.214	2.7609	-	valid
13	30.002	2.5956	-	valid
14	20.264	2.6970	-	valid
15	3.421	0.1573	-	valid
16	2.689	0.1240	-	valid
17	2.589	0.0291	-	valid
18	3.043	0.1240	-	valid
19	2.668	0.1240	-	valid
20	2.888	0.1240	-	valid

2. Several boundaries | seed 4 | Score base

t	Angle (deg)	Regret (%)	Balanced (%)	Status
1	56.194	10.6196	-	valid
2	59.715	11.7731	-	valid
3	80.231	7.9995	-	valid
4	49.562	6.3474	-	valid
5	39.057	1.7962	-	valid
6	26.044	1.8978	-	valid
7	41.129	2.6500	-	valid
8	33.890	1.7962	-	valid
9	45.788	1.8619	-	valid
10	35.762	1.7037	-	valid
11	38.030	1.7883	-	valid
12	37.248	1.7035	-	valid
13	36.024	1.7197	-	valid
14	42.362	1.7037	-	valid
15	40.313	1.7197	-	valid
16	31.659	1.7197	-	valid
17	26.627	1.7197	-	valid
18	8.118	1.3773	-	valid
19	1.816	0.0000	-	valid
20	2.033	0.0194	-	valid

Complete step records | runs 21-24 of 80

3. Similar / varied queries | seed 0 | Pedro

t	Angle (deg)	Regret (%)	Balanced (%)	Status
1	75.591	11.7451	19.7555	valid
2	43.297	2.7270	3.5085	valid
3	43.297	2.9941	3.7465	valid
4	40.858	2.9941	4.2156	valid
5	41.288	2.9941	4.2156	valid
6	37.788	2.3584	3.3018	valid
7	31.971	1.2834	1.9044	valid
8	31.971	1.2834	1.9044	valid
9	34.872	1.8804	2.5959	valid
10	34.872	1.8804	2.5959	valid
11	34.872	1.8804	2.5959	valid
12	34.872	1.8804	2.5959	valid
13	34.872	1.8804	2.5959	valid
14	39.387	4.1386	6.2811	valid
15	39.387	4.1386	6.2811	valid
16	39.387	4.1368	6.2573	valid
17	39.387	4.1368	6.2573	valid
18	39.387	4.1368	6.1788	valid
19	39.387	4.1368	6.2573	valid
20	39.387	4.1368	6.2573	valid

3. Similar / varied queries | seed 1 | Pedro

t	Angle (deg)	Regret (%)	Balanced (%)	Status
1	70.122	41.6901	43.2964	valid
2	43.049	6.0754	9.2159	valid
3	34.003	3.3016	2.7460	valid
4	35.409	3.0336	2.6505	valid
5	35.733	3.0302	2.6505	valid
6	35.733	3.0302	2.6505	valid
7	35.733	3.0302	2.6505	valid
8	23.092	0.9116	1.1066	valid
9	23.092	0.9116	1.1066	valid
10	23.092	0.9116	1.1066	valid
11	23.092	0.9116	1.1066	valid
12	23.092	0.9116	1.1066	valid
13	23.092	0.9116	1.1066	valid
14	23.092	0.9116	1.1066	valid
15	23.092	0.9116	1.1066	valid
16	23.092	0.9116	1.1066	valid
17	23.092	0.9116	1.1066	valid
18	23.092	0.9116	1.1066	valid
19	23.092	0.9116	1.1066	valid
20	23.092	0.9116	1.1066	valid

3. Similar / varied queries | seed 0 | Score base

t	Angle (deg)	Regret (%)	Balanced (%)	Status
1	31.862	2.4127	2.4028	valid
2	14.796	0.7048	0.6760	valid
3	22.525	10.4347	9.8468	valid
4	14.790	0.9305	1.4992	valid
5	18.801	0.2585	0.9723	valid
6	7.451	0.7926	0.6604	valid
7	8.254	0.1925	0.5245	valid
8	9.327	0.2947	0.5104	valid
9	9.777	0.1467	0.4227	valid
10	9.366	0.1467	0.3515	valid
11	5.378	0.1053	0.0832	valid
12	3.352	0.0821	0.0832	valid
13	7.306	0.0319	0.0879	valid
14	7.072	0.0182	0.0193	valid
15	5.449	0.0526	0.0651	valid
16	5.674	0.0000	0.0193	valid
17	4.447	0.0372	0.0473	valid
18	5.713	0.0478	0.0473	valid
19	6.757	0.0603	0.0473	valid
20	6.185	0.0149	0.0193	valid

3. Similar / varied queries | seed 1 | Score base

t	Angle (deg)	Regret (%)	Balanced (%)	Status
1	54.446	33.1835	32.9577	valid
2	40.987	5.2078	4.1433	valid
3	28.602	2.8649	3.5499	valid
4	28.758	2.7269	2.7164	valid
5	25.014	1.4403	2.4205	valid
6	7.609	0.3421	0.4223	valid
7	8.442	0.2255	0.3721	valid
8	8.454	0.2983	0.4798	valid
9	8.360	0.1875	0.4426	valid
10	5.902	0.0910	0.2028	valid
11	6.485	0.2983	0.4360	valid
12	6.003	0.7488	0.8464	valid
13	4.506	0.2983	0.3213	valid
14	5.913	0.1674	0.1982	valid
15	5.765	0.2118	0.3038	valid
16	5.569	0.1111	0.1178	valid
17	4.797	0.1148	0.1563	valid
18	5.411	0.1324	0.1519	valid
19	3.652	0.1090	0.1290	valid
20	7.111	0.1061	0.1982	valid

Complete step records | runs 25-28 of 80

3. Similar / varied queries | seed 2 | Pedro

t	Angle (deg)	Regret (%)	Balanced (%)	Status
1	45.808	35.5773	31.8158	valid
2	33.239	2.6512	3.8369	valid
3	39.309	4.2273	6.5048	valid
4	39.309	4.2273	6.5048	valid
5	37.645	3.5907	4.4653	valid
6	36.449	3.1671	4.3207	valid
7	30.307	1.9797	2.2452	valid
8	30.307	1.9797	2.2452	valid
9	30.307	1.9797	2.2452	valid
10	30.307	1.9797	2.2452	valid
11	22.654	1.8348	1.1493	valid
12	22.654	1.8348	1.1493	valid
13	22.654	1.8348	1.1493	valid
14	22.654	1.8348	1.1493	valid
15	22.654	1.8348	1.1493	valid
16	22.654	1.8348	1.1493	valid
17	22.654	1.8348	1.1493	valid
18	23.155	1.8348	1.1493	valid
19	23.155	1.8348	1.1493	valid
20	23.155	1.8348	1.1493	valid

3. Similar / varied queries | seed 3 | Pedro

t	Angle (deg)	Regret (%)	Balanced (%)	Status
1	75.576	11.5573	19.7780	valid
2	43.140	2.6302	3.3324	valid
3	43.140	2.6302	3.4758	valid
4	23.432	0.6887	0.8268	valid
5	23.971	0.6887	0.8268	valid
6	40.059	2.3087	3.4831	valid
7	40.059	2.3087	3.4831	valid
8	41.994	1.7565	3.1576	valid
9	41.994	1.7565	3.1576	valid
10	41.994	1.7565	3.1576	valid
11	41.994	1.7565	3.1576	valid
12	41.994	1.7565	3.1576	valid
13	41.994	1.7565	3.1576	valid
14	41.994	1.7565	3.1576	valid
15	41.994	1.7565	3.1576	valid
16	41.994	1.7565	3.1576	valid
17	41.994	1.7565	3.1576	valid
18	41.994	1.7565	3.1576	valid
19	41.148	1.7561	2.7292	valid
20	41.148	1.7561	2.7292	valid

3. Similar / varied queries | seed 2 | Score base

t	Angle (deg)	Regret (%)	Balanced (%)	Status
1	18.289	3.4579	3.0089	valid
2	21.497	5.1140	4.1492	valid
3	18.810	4.4471	4.2386	valid
4	12.100	0.9172	0.8765	valid
5	13.631	0.5180	0.6487	valid
6	8.468	0.3108	0.2745	valid
7	3.385	0.3314	0.3733	valid
8	4.557	0.4194	0.4336	valid
9	2.607	0.2996	0.1520	valid
10	4.767	0.0149	0.0876	valid
11	8.009	0.0407	0.3076	valid
12	5.851	0.2198	0.1145	valid
13	1.824	0.1672	0.1119	valid
14	4.542	0.2348	0.1831	valid
15	4.443	0.0959	0.0831	valid
16	2.686	0.0992	0.0597	valid
17	2.475	0.0959	0.0448	valid
18	4.501	0.1170	0.0448	valid
19	4.080	0.0407	0.0284	valid
20	2.344	0.0321	0.0329	valid

3. Similar / varied queries | seed 3 | Score base

t	Angle (deg)	Regret (%)	Balanced (%)	Status
1	84.048	67.3372	65.7054	valid
2	38.557	13.5098	14.5532	valid
3	47.826	5.3238	8.2324	valid
4	21.066	0.8823	1.5086	valid
5	7.724	0.1583	0.5605	valid
6	15.526	0.6258	0.4258	valid
7	13.383	0.7442	0.9115	valid
8	14.128	0.7869	0.4727	valid
9	13.982	0.6132	0.5213	valid
10	19.520	0.6674	1.4283	valid
11	15.782	0.6449	1.0302	valid
12	12.650	0.4181	0.4287	valid
13	14.167	0.6449	0.5268	valid
14	9.311	0.1864	0.3867	valid
15	5.167	0.0747	0.1736	valid
16	4.753	0.1150	0.3854	valid
17	4.469	0.1371	0.4108	valid
18	3.142	0.1150	0.3812	valid
19	2.822	0.0286	0.0183	valid
20	1.602	0.0012	0.0042	valid

Complete step records | runs 29-32 of 80

3. Similar / varied queries | seed 4 | Pedro

t	Angle (deg)	Regret (%)	Balanced (%)	Status
1	34.052	1.2945	3.3277	valid
2	34.593	1.2945	3.3277	valid
3	42.990	3.0374	5.7265	valid
4	42.226	2.7897	4.8366	valid
5	42.199	2.7897	4.8366	valid
6	42.199	2.7897	4.8366	valid
7	40.298	2.6481	4.8366	valid
8	31.918	1.9589	2.6605	valid
9	31.918	1.9589	2.6605	valid
10	24.411	0.7507	2.0720	valid
11	24.411	0.7507	2.0720	valid
12	24.996	0.8555	2.0720	valid
13	24.996	0.8555	2.0720	valid
14	25.049	0.8555	2.0720	valid
15	26.220	0.8815	2.0720	valid
16	26.317	0.7507	2.1373	valid
17	26.391	0.7507	2.1373	valid
18	26.045	0.7507	2.1373	valid
19	26.045	0.7507	2.1373	valid
20	26.045	0.7507	2.1373	valid

4. Balanced choice | seed 0 | Pedro

t	Angle (deg)	Regret (%)	Balanced (%)	Status
1	43.016	7.8862	-	valid
2	35.197	6.5878	-	valid
3	35.197	6.5878	-	valid
4	9.625	0.8931	-	valid
5	9.625	0.8931	-	valid
6	7.923	0.6626	-	valid
7	7.923	0.6626	-	valid
8	7.923	0.6626	-	valid
9	7.923	0.6626	-	valid
10	7.923	0.6626	-	valid
11	7.923	0.6626	-	valid
12	7.923	0.6626	-	valid
13	7.923	0.6626	-	valid
14	7.923	0.6626	-	valid
15	7.923	0.6626	-	valid
16	7.923	0.6626	-	valid
17	7.923	0.6626	-	valid
18	7.923	0.6626	-	valid
19	7.923	0.6626	-	valid
20	7.923	0.6626	-	valid

3. Similar / varied queries | seed 4 | Score base

t	Angle (deg)	Regret (%)	Balanced (%)	Status
1	65.200	28.8514	29.9477	valid
2	23.817	1.0056	1.9666	valid
3	18.077	3.6710	2.7043	valid
4	26.394	0.9693	2.4086	valid
5	19.179	0.2568	0.3966	valid
6	13.167	0.2918	0.1906	valid
7	9.581	0.4418	0.2059	valid
8	7.448	0.1602	0.0463	valid
9	12.493	1.5914	1.3544	valid
10	6.109	0.1078	0.0803	valid
11	7.649	0.1384	0.0890	valid
12	9.879	0.1730	0.2035	valid
13	7.987	0.1350	0.0891	valid
14	11.639	0.0837	0.1180	valid
15	9.134	0.1223	0.1304	valid
16	6.015	0.0557	0.0379	valid
17	9.919	0.0919	0.0822	valid
18	8.452	0.0887	0.0757	valid
19	8.485	0.0837	0.0529	valid
20	9.831	0.0919	0.0820	valid

4. Balanced choice | seed 0 | Score base

t	Angle (deg)	Regret (%)	Balanced (%)	Status
1	57.862	31.4900	-	valid
2	55.474	32.6131	-	valid
3	6.848	0.4335	-	valid
4	11.540	1.1461	-	valid
5	4.712	0.2858	-	valid
6	9.421	0.5682	-	valid
7	5.065	0.3181	-	valid
8	6.924	0.3809	-	valid
9	5.379	0.3181	-	valid
10	1.360	0.0271	-	valid
11	2.945	0.1983	-	valid
12	5.568	0.3604	-	valid
13	2.545	0.0380	-	valid
14	1.196	0.0000	-	valid
15	6.594	0.3292	-	valid
16	5.327	0.3604	-	valid
17	1.158	0.0115	-	valid
18	2.511	0.0892	-	valid
19	1.861	0.0000	-	valid
20	1.308	0.0946	-	valid

Complete step records | runs 33-36 of 80

4. Balanced choice | seed 1 | Pedro

t	Angle (deg)	Regret (%)	Balanced (%)	Status
1	76.919	59.2947	-	valid
2	22.519	2.0739	-	valid
3	34.130	11.9738	-	valid
4	34.130	11.9738	-	valid
5	41.220	19.7685	-	valid
6	38.449	15.7221	-	valid
7	38.449	15.7221	-	valid
8	38.505	15.9629	-	valid
9	14.951	1.3467	-	valid
10	14.951	1.3467	-	valid
11	14.951	1.3467	-	valid
12	14.951	1.3467	-	valid
13	14.951	1.3467	-	valid
14	14.951	1.3467	-	valid
15	14.951	1.3467	-	valid
16	20.837	2.5705	-	valid
17	20.837	2.5705	-	valid
18	20.837	2.5705	-	valid
19	20.837	2.5705	-	valid
20	20.837	2.5705	-	valid

4. Balanced choice | seed 2 | Pedro

t	Angle (deg)	Regret (%)	Balanced (%)	Status
1	60.453	25.8292	-	valid
2	33.129	4.3752	-	valid
3	36.685	7.1805	-	valid
4	32.927	7.7417	-	valid
5	33.373	7.7417	-	valid
6	33.373	7.7417	-	valid
7	21.282	2.9956	-	valid
8	21.282	2.9956	-	valid
9	21.282	2.9956	-	valid
10	21.282	2.9956	-	valid
11	21.282	2.9956	-	valid
12	21.282	2.9956	-	valid
13	21.282	2.9956	-	valid
14	6.382	0.3015	-	valid
15	6.382	0.3015	-	valid
16	6.382	0.3015	-	valid
17	6.382	0.3015	-	valid
18	6.382	0.3015	-	valid
19	6.382	0.3015	-	valid
20	6.382	0.3015	-	valid

4. Balanced choice | seed 1 | Score base

t	Angle (deg)	Regret (%)	Balanced (%)	Status
1	63.661	38.2588	-	valid
2	34.467	7.7085	-	valid
3	22.611	5.7869	-	valid
4	14.900	1.5448	-	valid
5	9.641	0.8047	-	valid
6	5.435	0.4191	-	valid
7	7.924	0.4155	-	valid
8	7.198	0.5009	-	valid
9	2.703	0.0558	-	valid
10	6.590	0.4664	-	valid
11	8.622	0.5486	-	valid
12	4.700	0.0815	-	valid
13	8.210	0.6022	-	valid
14	5.508	0.1112	-	valid
15	4.270	0.1178	-	valid
16	2.296	0.0294	-	valid
17	3.366	0.1101	-	valid
18	2.839	0.0657	-	valid
19	2.550	0.0982	-	valid
20	2.043	0.0000	-	valid

4. Balanced choice | seed 2 | Score base

t	Angle (deg)	Regret (%)	Balanced (%)	Status
1	86.154	60.4161	-	valid
2	74.596	43.3580	-	valid
3	36.731	12.0802	-	valid
4	14.402	1.2506	-	valid
5	10.058	0.8792	-	valid
6	3.957	0.0689	-	valid
7	6.448	0.6223	-	valid
8	3.469	0.2127	-	valid
9	3.180	0.0887	-	valid
10	5.590	0.1387	-	valid
11	4.342	0.2240	-	valid
12	1.578	0.0308	-	valid
13	2.771	0.0581	-	valid
14	2.966	0.0771	-	valid
15	3.033	0.0952	-	valid
16	4.289	0.2304	-	valid
17	3.190	0.0952	-	valid
18	2.630	0.0771	-	valid
19	3.934	0.1495	-	valid
20	3.376	0.0771	-	valid

Complete step records | runs 37-40 of 80

4. Balanced choice | seed 3 | Pedro

t	Angle (deg)	Regret (%)	Balanced (%)	Status
1	46.067	9.1259	-	valid
2	29.993	3.2335	-	valid
3	11.936	1.2699	-	valid
4	26.368	3.9823	-	valid
5	30.006	6.1538	-	valid
6	30.006	6.1538	-	valid
7	18.575	3.0839	-	valid
8	10.450	0.5109	-	valid
9	10.450	0.5109	-	valid
10	11.540	0.7064	-	valid
11	11.540	0.7064	-	valid
12	11.540	0.7064	-	valid
13	11.540	0.7064	-	valid
14	11.540	0.7064	-	valid
15	11.540	0.7064	-	valid
16	11.540	0.7064	-	valid
17	11.532	1.0970	-	valid
18	11.865	1.1891	-	valid
19	11.865	1.1891	-	valid
20	11.865	1.1891	-	valid

4. Balanced choice | seed 4 | Pedro

t	Angle (deg)	Regret (%)	Balanced (%)	Status
1	35.264	10.3284	-	valid
2	53.944	23.0108	-	valid
3	47.902	21.0013	-	valid
4	47.902	21.0013	-	valid
5	44.547	18.1474	-	valid
6	44.547	18.1474	-	valid
7	44.547	18.1474	-	valid
8	44.547	18.1474	-	valid
9	10.113	0.5725	-	valid
10	10.113	0.5725	-	valid
11	12.096	0.8497	-	valid
12	12.096	0.8497	-	valid
13	12.096	0.8497	-	valid
14	12.096	0.8497	-	valid
15	12.096	0.8497	-	valid
16	12.096	0.8497	-	valid
17	12.096	0.8497	-	valid
18	12.096	0.8497	-	valid
19	6.196	0.3991	-	valid
20	6.362	0.3126	-	valid

4. Balanced choice | seed 3 | Score base

t	Angle (deg)	Regret (%)	Balanced (%)	Status
1	52.600	9.5988	-	valid
2	61.776	36.5943	-	valid
3	41.015	20.4660	-	valid
4	18.512	1.9650	-	valid
5	11.464	1.4955	-	valid
6	9.027	0.4569	-	valid
7	12.885	1.3756	-	valid
8	5.637	0.2967	-	valid
9	11.283	1.1293	-	valid
10	7.210	0.4099	-	valid
11	9.736	0.4688	-	valid
12	5.726	0.2660	-	valid
13	7.504	0.3072	-	valid
14	7.316	0.3072	-	valid
15	6.082	0.1859	-	valid
16	6.001	0.2811	-	valid
17	4.041	0.1598	-	valid
18	3.083	0.0256	-	valid
19	2.833	0.1007	-	valid
20	1.356	0.0105	-	valid

4. Balanced choice | seed 4 | Score base

t	Angle (deg)	Regret (%)	Balanced (%)	Status
1	37.997	12.1777	-	valid
2	35.344	7.9782	-	valid
3	7.657	0.2117	-	valid
4	5.719	0.3263	-	valid
5	8.108	0.7200	-	valid
6	7.457	0.5798	-	valid
7	6.825	0.3799	-	valid
8	8.737	0.4896	-	valid
9	8.759	0.4474	-	valid
10	7.951	0.3482	-	valid
11	6.870	0.3029	-	valid
12	6.096	0.2738	-	valid
13	7.439	0.2734	-	valid
14	5.253	0.2317	-	valid
15	3.779	0.0635	-	valid
16	3.547	0.0823	-	valid
17	2.934	0.0656	-	valid
18	2.372	0.0001	-	valid
19	1.954	0.0001	-	valid
20	1.956	0.0029	-	valid

Complete step records | runs 41-44 of 80

5. Subset selection | seed 0 | Pedro

t	Angle (deg)	Regret (%)	Balanced (%)	Status
1	38.702	16.6875	-	valid
2	43.454	18.4411	-	valid
3	22.427	4.3769	-	valid
4	20.661	2.4256	-	valid
5	13.539	0.4121	-	valid
6	13.539	0.4121	-	valid
7	13.539	0.4121	-	valid
8	15.363	0.4853	-	valid
9	15.363	0.4853	-	valid
10	15.363	0.4853	-	valid
11	20.491	1.1756	-	valid
12	17.365	0.8449	-	valid
13	12.955	0.3895	-	valid
14	12.955	0.3895	-	valid
15	12.955	0.3895	-	valid
16	12.861	0.3895	-	valid
17	4.880	0.0625	-	valid
18	4.880	0.0625	-	valid
19	4.880	0.0625	-	valid
20	4.880	0.0625	-	valid

5. Subset selection | seed 1 | Pedro

t	Angle (deg)	Regret (%)	Balanced (%)	Status
1	38.961	19.7104	-	valid
2	10.683	0.3094	-	valid
3	10.683	0.3094	-	valid
4	7.819	0.1023	-	valid
5	7.819	0.1023	-	valid
6	4.722	0.0796	-	valid
7	11.649	0.4498	-	valid
8	11.649	0.4498	-	valid
9	11.649	0.4498	-	valid
10	11.957	0.3104	-	valid
11	11.957	0.3104	-	valid
12	11.957	0.3104	-	valid
13	13.476	0.5108	-	valid
14	13.476	0.5108	-	valid
15	13.476	0.5108	-	valid
16	13.476	0.5108	-	valid
17	13.476	0.5108	-	valid
18	13.476	0.5108	-	valid
19	8.522	0.1315	-	valid
20	8.522	0.1315	-	valid

5. Subset selection | seed 0 | Score base

t	Angle (deg)	Regret (%)	Balanced (%)	Status
1	9.920	0.3184	-	valid
2	16.438	0.6808	-	valid
3	13.801	0.4883	-	valid
4	6.048	0.0283	-	valid
5	9.490	0.1695	-	valid
6	8.108	0.1075	-	valid
7	9.104	0.1449	-	valid
8	9.198	0.1785	-	valid
9	11.208	0.2179	-	valid
10	6.406	0.0075	-	valid
11	6.470	0.0292	-	valid
12	7.072	0.0675	-	valid
13	4.187	0.0366	-	valid
14	3.819	0.0444	-	valid
15	3.563	0.0268	-	valid
16	3.860	0.0075	-	valid
17	5.790	0.0344	-	valid
18	5.348	0.0366	-	valid
19	3.452	0.0075	-	valid
20	5.090	0.0075	-	valid

5. Subset selection | seed 1 | Score base

t	Angle (deg)	Regret (%)	Balanced (%)	Status
1	38.387	13.0124	-	valid
2	25.086	4.4168	-	valid
3	21.729	4.5923	-	valid
4	14.776	0.2895	-	valid
5	8.621	0.0713	-	valid
6	9.473	0.1693	-	valid
7	4.808	0.0213	-	valid
8	5.949	0.0289	-	valid
9	7.585	0.0775	-	valid
10	6.782	0.1304	-	valid
11	6.441	0.0819	-	valid
12	6.020	0.0450	-	valid
13	6.369	0.1222	-	valid
14	9.374	0.1415	-	valid
15	5.922	0.0450	-	valid
16	3.567	0.0132	-	valid
17	2.685	0.0247	-	valid
18	3.460	0.0057	-	valid
19	1.754	0.0057	-	valid
20	1.740	0.0068	-	valid

Complete step records | runs 45-48 of 80

5. Subset selection | seed 2 | Pedro

t	Angle (deg)	Regret (%)	Balanced (%)	Status
1	16.000	0.4953	-	valid
2	15.367	0.9808	-	valid
3	18.823	0.8780	-	valid
4	18.430	0.8780	-	valid
5	19.781	2.4883	-	valid
6	19.781	2.4883	-	valid
7	14.100	0.5012	-	valid
8	14.100	0.5012	-	valid
9	12.299	0.3935	-	valid
10	9.809	0.1991	-	valid
11	9.809	0.1991	-	valid
12	14.499	0.3007	-	valid
13	14.499	0.3007	-	valid
14	14.499	0.3007	-	valid
15	14.499	0.3007	-	valid
16	14.499	0.3007	-	valid
17	14.499	0.3007	-	valid
18	14.499	0.3007	-	valid
19	14.499	0.3007	-	valid
20	14.499	0.3007	-	valid

5. Subset selection | seed 2 | Score base

t	Angle (deg)	Regret (%)	Balanced (%)	Status
1	32.514	5.6711	-	valid
2	29.708	3.2694	-	valid
3	10.093	0.2870	-	valid
4	9.859	0.3258	-	valid
5	7.033	0.2690	-	valid
6	12.432	0.7061	-	valid
7	10.775	0.2278	-	valid
8	5.538	0.0643	-	valid
9	6.282	0.1398	-	valid
10	5.918	0.0995	-	valid
11	5.300	0.0643	-	valid
12	6.640	0.1721	-	valid
13	5.692	0.0720	-	valid
14	5.550	0.0995	-	valid
15	2.420	0.0180	-	valid
16	3.946	0.0988	-	valid
17	4.749	0.0631	-	valid
18	3.440	0.0180	-	valid
19	2.635	0.0156	-	valid
20	3.966	0.0631	-	valid

5. Subset selection | seed 3 | Pedro

t	Angle (deg)	Regret (%)	Balanced (%)	Status
1	34.600	14.1374	-	valid
2	36.371	14.9397	-	valid
3	17.466	2.9699	-	valid
4	12.911	0.2182	-	valid
5	15.958	0.2654	-	valid
6	14.042	0.3046	-	valid
7	14.042	0.3046	-	valid
8	11.843	0.4381	-	valid
9	11.843	0.4381	-	valid
10	12.085	0.4448	-	valid
11	3.220	0.0295	-	valid
12	3.220	0.0295	-	valid
13	3.220	0.0295	-	valid
14	6.912	0.1337	-	valid
15	6.912	0.1337	-	valid
16	6.912	0.1337	-	valid
17	6.912	0.1337	-	valid
18	6.912	0.1337	-	valid
19	6.912	0.1337	-	valid
20	6.912	0.1337	-	valid

5. Subset selection | seed 3 | Score base

t	Angle (deg)	Regret (%)	Balanced (%)	Status
1	41.086	13.5714	-	valid
2	18.516	0.4694	-	valid
3	19.724	0.8824	-	valid
4	12.229	0.2981	-	valid
5	10.899	0.2424	-	valid
6	10.268	0.1548	-	valid
7	6.951	0.0802	-	valid
8	6.815	0.1013	-	valid
9	8.941	0.1819	-	valid
10	7.026	0.1065	-	valid
11	6.354	0.0594	-	valid
12	6.712	0.0811	-	valid
13	5.280	0.0528	-	valid
14	5.254	0.0879	-	valid
15	4.559	0.0082	-	valid
16	5.509	0.0655	-	valid
17	5.543	0.0447	-	valid
18	4.968	0.0617	-	valid
19	3.615	0.0295	-	valid
20	3.347	0.0439	-	valid

Complete step records | runs 49-52 of 80

5. Subset selection | seed 4 | Pedro

t	Angle (deg)	Regret (%)	Balanced (%)	Status
1	17.685	0.4645	-	valid
2	6.709	0.0503	-	valid
3	19.211	1.6994	-	valid
4	19.211	1.6994	-	valid
5	28.531	8.3980	-	valid
6	28.531	8.3980	-	valid
7	28.531	8.3980	-	valid
8	28.531	8.3980	-	valid
9	28.531	8.3980	-	valid
10	24.344	4.1668	-	valid
11	24.344	4.1668	-	valid
12	16.205	0.5024	-	valid
13	16.205	0.5024	-	valid
14	16.205	0.5024	-	valid
15	16.205	0.5024	-	valid
16	16.205	0.5024	-	valid
17	16.205	0.5024	-	valid
18	10.369	0.2373	-	valid
19	10.369	0.2373	-	valid
20	10.369	0.2373	-	valid

6. Budget selection | seed 0 | Pedro

t	Angle (deg)	Regret (%)	Balanced (%)	Status
1	25.595	6.5948	-	valid
2	16.888	1.5064	-	valid
3	17.531	1.6147	-	valid
4	12.425	0.9482	-	valid
5	22.323	1.9119	-	valid
6	22.323	1.9119	-	valid
7	21.028	1.8361	-	valid
8	20.920	1.8361	-	valid
9	20.920	1.8361	-	valid
10	20.920	1.8361	-	valid
11	20.920	1.8361	-	valid
12	22.811	2.2782	-	valid
13	22.811	2.2782	-	valid
14	15.559	0.8819	-	valid
15	15.559	0.8819	-	valid
16	15.559	0.8819	-	valid
17	20.440	1.4285	-	valid
18	20.440	1.4285	-	valid
19	20.440	1.4285	-	valid
20	20.440	1.4285	-	valid

5. Subset selection | seed 4 | Score base

t	Angle (deg)	Regret (%)	Balanced (%)	Status
1	40.619	16.7017	-	valid
2	17.572	0.7425	-	valid
3	8.348	0.2251	-	valid
4	6.286	0.1343	-	valid
5	6.029	0.0429	-	valid
6	6.228	0.0404	-	valid
7	3.298	0.0139	-	valid
8	4.653	0.0273	-	valid
9	3.789	0.0145	-	valid
10	2.186	0.0000	-	valid
11	2.851	0.0195	-	valid
12	3.404	0.0148	-	valid
13	4.696	0.0560	-	valid
14	4.847	0.0490	-	valid
15	1.394	0.0064	-	valid
16	3.130	0.0195	-	valid
17	1.968	0.0059	-	valid
18	1.133	0.0093	-	valid
19	3.682	0.0896	-	valid
20	2.203	0.0395	-	valid

6. Budget selection | seed 0 | Score base

t	Angle (deg)	Regret (%)	Balanced (%)	Status
1	48.780	12.5918	-	valid
2	27.843	3.3549	-	valid
3	22.615	2.6849	-	valid
4	13.988	0.7515	-	valid
5	8.412	0.1034	-	valid
6	8.700	0.0703	-	valid
7	9.254	0.1075	-	valid
8	6.495	0.1798	-	valid
9	5.882	0.0704	-	valid
10	6.298	0.1881	-	valid
11	6.958	0.0643	-	valid
12	8.783	0.3515	-	valid
13	9.433	0.1408	-	valid
14	7.699	0.0665	-	valid
15	6.829	0.0746	-	valid
16	7.233	0.0852	-	valid
17	8.406	0.0973	-	valid
18	8.273	0.1136	-	valid
19	8.878	0.0746	-	valid
20	8.342	0.0807	-	valid

Complete step records | runs 53-56 of 80

6. Budget selection | seed 1 | Pedro

t	Angle (deg)	Regret (%)	Balanced (%)	Status
1	28.533	6.2375	-	valid
2	28.533	6.2375	-	valid
3	31.026	6.1486	-	valid
4	31.026	6.1486	-	valid
5	31.026	6.1486	-	valid
6	31.026	6.1486	-	valid
7	25.655	1.9945	-	valid
8	16.266	0.9127	-	valid
9	17.241	0.9670	-	valid
10	30.222	1.8780	-	valid
11	23.927	1.3547	-	valid
12	23.927	1.3547	-	valid
13	23.927	1.3547	-	valid
14	24.060	1.3547	-	valid
15	24.060	1.3547	-	valid
16	24.060	1.3547	-	valid
17	14.267	0.2685	-	valid
18	14.267	0.2685	-	valid
19	13.507	0.2394	-	valid
20	13.507	0.2394	-	valid

6. Budget selection | seed 2 | Pedro

t	Angle (deg)	Regret (%)	Balanced (%)	Status
1	14.444	1.3540	-	valid
2	19.961	1.8654	-	valid
3	27.725	2.3930	-	valid
4	27.644	2.3930	-	valid
5	28.251	3.8487	-	valid
6	25.284	2.0324	-	valid
7	23.164	1.9571	-	valid
8	23.164	1.9571	-	valid
9	23.857	1.9571	-	valid
10	18.640	2.1639	-	valid
11	19.827	2.5508	-	valid
12	19.827	2.5508	-	valid
13	19.827	2.5508	-	valid
14	19.827	2.5508	-	valid
15	19.827	2.5508	-	valid
16	19.827	2.5508	-	valid
17	19.827	2.5508	-	valid
18	19.827	2.5508	-	valid
19	19.827	2.5508	-	valid
20	19.827	2.5508	-	valid

6. Budget selection | seed 1 | Score base

t	Angle (deg)	Regret (%)	Balanced (%)	Status
1	38.194	16.5762	-	valid
2	10.893	0.1206	-	valid
3	19.080	2.2914	-	valid
4	17.265	0.7738	-	valid
5	14.649	0.6534	-	valid
6	7.221	0.1180	-	valid
7	13.563	0.3063	-	valid
8	14.959	0.6780	-	valid
9	15.387	0.7804	-	valid
10	11.691	0.1872	-	valid
11	8.490	0.1874	-	valid
12	9.540	0.0905	-	valid
13	9.723	0.2734	-	valid
14	3.900	0.0253	-	valid
15	4.632	0.0087	-	valid
16	5.057	0.0496	-	valid
17	8.017	0.1371	-	valid
18	3.595	0.0087	-	valid
19	6.261	0.0447	-	valid
20	2.391	0.0238	-	valid

6. Budget selection | seed 2 | Score base

t	Angle (deg)	Regret (%)	Balanced (%)	Status
1	55.916	31.2006	-	valid
2	28.441	8.8633	-	valid
3	17.926	1.7570	-	valid
4	8.898	0.2113	-	valid
5	11.784	0.4161	-	valid
6	6.170	0.0665	-	valid
7	7.134	0.0972	-	valid
8	6.142	0.0296	-	valid
9	5.617	0.0536	-	valid
10	5.034	0.0742	-	valid
11	4.919	0.0433	-	valid
12	5.412	0.0536	-	valid
13	8.163	0.1657	-	valid
14	6.039	0.0998	-	valid
15	3.189	0.0296	-	valid
16	5.318	0.0630	-	valid
17	5.441	0.0768	-	valid
18	8.379	0.1657	-	valid
19	6.139	0.0536	-	valid
20	5.715	0.0536	-	valid

Complete step records | runs 57-60 of 80

6. Budget selection | seed 3 | Pedro

t	Angle (deg)	Regret (%)	Balanced (%)	Status
1	29.064	7.9521	-	valid
2	24.850	3.3191	-	valid
3	29.169	3.7618	-	valid
4	29.233	7.5522	-	valid
5	29.358	7.6408	-	valid
6	29.321	7.4666	-	valid
7	29.321	7.4666	-	valid
8	29.000	3.5876	-	valid
9	29.000	3.5876	-	valid
10	13.289	0.6317	-	valid
11	11.978	0.1327	-	valid
12	11.978	0.1327	-	valid
13	10.899	0.0869	-	valid
14	10.899	0.0869	-	valid
15	10.899	0.0869	-	valid
16	10.638	0.2133	-	valid
17	10.664	0.2133	-	valid
18	10.664	0.2133	-	valid
19	10.962	0.1918	-	valid
20	10.538	0.1918	-	valid

6. Budget selection | seed 4 | Pedro

t	Angle (deg)	Regret (%)	Balanced (%)	Status
1	50.558	25.1836	-	valid
2	25.820	3.3779	-	valid
3	35.700	4.8370	-	valid
4	35.015	8.3911	-	valid
5	20.731	1.5007	-	valid
6	20.731	1.5007	-	valid
7	20.731	1.5007	-	valid
8	15.318	0.8985	-	valid
9	15.318	0.8985	-	valid
10	15.318	0.8985	-	valid
11	15.318	0.8985	-	valid
12	15.318	0.8985	-	valid
13	15.318	0.8985	-	valid
14	15.318	0.8985	-	valid
15	15.318	0.8985	-	valid
16	15.318	0.8985	-	valid
17	15.116	0.8985	-	valid
18	15.116	0.8985	-	valid
19	15.116	0.8985	-	valid
20	16.956	0.8313	-	valid

6. Budget selection | seed 3 | Score base

t	Angle (deg)	Regret (%)	Balanced (%)	Status
1	8.195	0.1019	-	valid
2	21.618	2.6089	-	valid
3	8.605	0.1119	-	valid
4	15.357	1.3217	-	valid
5	13.893	0.5582	-	valid
6	9.978	0.3652	-	valid
7	6.030	0.1077	-	valid
8	6.536	0.1914	-	valid
9	6.943	0.0964	-	valid
10	8.687	0.4326	-	valid
11	8.487	0.1514	-	valid
12	6.106	0.1219	-	valid
13	9.580	0.3340	-	valid
14	7.589	0.1464	-	valid
15	6.065	0.0964	-	valid
16	4.382	0.0362	-	valid
17	1.557	0.0000	-	valid
18	4.785	0.0647	-	valid
19	6.803	0.0958	-	valid
20	7.261	0.0362	-	valid

6. Budget selection | seed 4 | Score base

t	Angle (deg)	Regret (%)	Balanced (%)	Status
1	44.048	17.0839	-	valid
2	39.868	19.1969	-	valid
3	12.760	0.5899	-	valid
4	11.813	0.2545	-	valid
5	6.689	0.0494	-	valid
6	11.280	0.4566	-	valid
7	8.138	0.2310	-	valid
8	8.780	0.1636	-	valid
9	7.755	0.2176	-	valid
10	4.545	0.0540	-	valid
11	7.270	0.0752	-	valid
12	5.270	0.0000	-	valid
13	6.026	0.0000	-	valid
14	5.528	0.0494	-	valid
15	4.864	0.0669	-	valid
16	6.621	0.0704	-	valid
17	4.457	0.0446	-	valid
18	4.718	0.0561	-	valid
19	6.071	0.1198	-	valid
20	3.753	0.0752	-	valid

Complete step records | runs 61-64 of 80

7. Stronger noise | seed 0 | Pedro

t	Angle (deg)	Regret (%)	Balanced (%)	Status
1	43.016	7.8862	-	valid
2	35.197	6.5878	-	valid
3	35.197	6.5878	-	valid
4	33.486	6.1289	-	valid
5	33.486	6.1289	-	valid
6	31.298	7.4432	-	valid
7	31.298	7.4432	-	valid
8	31.298	7.4432	-	valid
9	31.298	7.4432	-	valid
10	31.298	7.4432	-	valid
11	31.298	7.4432	-	valid
12	31.298	7.4432	-	valid
13	16.599	1.7693	-	valid
14	16.599	1.7693	-	valid
15	16.591	1.7693	-	valid
16	16.320	1.7693	-	valid
17	16.253	1.7815	-	valid
18	16.253	1.7815	-	valid
19	16.253	1.7815	-	valid
20	16.253	1.7815	-	valid

7. Stronger noise | seed 1 | Pedro

t	Angle (deg)	Regret (%)	Balanced (%)	Status
1	76.919	59.2947	-	valid
2	22.519	2.0739	-	valid
3	34.130	11.9738	-	valid
4	34.130	11.9738	-	valid
5	41.220	19.7685	-	valid
6	38.449	15.7221	-	valid
7	38.449	15.7221	-	valid
8	38.505	15.9629	-	valid
9	14.951	1.3467	-	valid
10	14.951	1.3467	-	valid
11	14.951	1.3467	-	valid
12	14.951	1.3467	-	valid
13	14.951	1.3467	-	valid
14	14.951	1.3467	-	valid
15	14.951	1.3467	-	valid
16	20.837	2.5705	-	valid
17	20.837	2.5705	-	valid
18	20.837	2.5705	-	valid
19	20.837	2.5705	-	valid
20	20.837	2.5705	-	valid

7. Stronger noise | seed 0 | Score base

t	Angle (deg)	Regret (%)	Balanced (%)	Status
1	57.862	31.4900	-	valid
2	55.474	32.6131	-	valid
3	6.848	0.4335	-	valid
4	24.862	5.2890	-	valid
5	8.971	1.0961	-	valid
6	10.061	0.6149	-	valid
7	7.727	0.6648	-	valid
8	9.090	0.4689	-	valid
9	9.390	0.5035	-	valid
10	10.365	0.7425	-	valid
11	8.847	0.6594	-	valid
12	9.123	0.7425	-	valid
13	2.722	0.0892	-	valid
14	7.220	0.4034	-	valid
15	6.875	0.4034	-	valid
16	5.401	0.2885	-	valid
17	4.068	0.0214	-	valid
18	6.027	0.2880	-	valid
19	6.144	0.2318	-	valid
20	6.183	0.3199	-	valid

7. Stronger noise | seed 1 | Score base

t	Angle (deg)	Regret (%)	Balanced (%)	Status
1	63.661	38.2588	-	valid
2	34.467	7.7085	-	valid
3	22.611	5.7869	-	valid
4	14.900	1.5448	-	valid
5	9.641	0.8047	-	valid
6	5.435	0.4191	-	valid
7	7.924	0.4155	-	valid
8	7.198	0.5009	-	valid
9	9.710	1.0409	-	valid
10	3.841	0.0815	-	valid
11	6.262	0.2470	-	valid
12	9.249	0.7844	-	valid
13	5.369	0.1678	-	valid
14	9.058	0.6809	-	valid
15	7.237	0.5161	-	valid
16	6.492	0.4609	-	valid
17	9.383	0.4609	-	valid
18	6.197	0.4664	-	valid
19	5.026	0.1623	-	valid
20	3.767	0.1178	-	valid

Complete step records | runs 65-68 of 80

7. Stronger noise | seed 2 | Pedro

t	Angle (deg)	Regret (%)	Balanced (%)	Status
1	60.453	25.8292	-	valid
2	33.129	4.3752	-	valid
3	36.685	7.1805	-	valid
4	32.927	7.7417	-	valid
5	33.373	7.7417	-	valid
6	33.373	7.7417	-	valid
7	21.282	2.9956	-	valid
8	21.282	2.9956	-	valid
9	21.282	2.9956	-	valid
10	21.282	2.9956	-	valid
11	21.282	2.9956	-	valid
12	21.282	2.9956	-	valid
13	21.282	2.9956	-	valid
14	21.282	2.9956	-	valid
15	21.282	2.9956	-	valid
16	21.282	2.9956	-	valid
17	21.282	2.9956	-	valid
18	21.282	2.9956	-	valid
19	21.282	2.9956	-	valid
20	21.282	2.9956	-	valid

7. Stronger noise | seed 3 | Pedro

t	Angle (deg)	Regret (%)	Balanced (%)	Status
1	46.067	9.1259	-	valid
2	29.993	3.2335	-	valid
3	11.936	1.2699	-	valid
4	26.368	3.9823	-	valid
5	30.006	6.1538	-	valid
6	30.006	6.1538	-	valid
7	18.575	3.0839	-	valid
8	10.450	0.5109	-	valid
9	10.450	0.5109	-	valid
10	11.540	0.7064	-	valid
11	11.540	0.7064	-	valid
12	11.540	0.7064	-	valid
13	11.540	0.7064	-	valid
14	11.540	0.7064	-	valid
15	11.540	0.7064	-	valid
16	11.540	0.7064	-	valid
17	11.532	1.0970	-	valid
18	11.865	1.1891	-	valid
19	11.865	1.1891	-	valid
20	11.865	1.1891	-	valid

7. Stronger noise | seed 2 | Score base

t	Angle (deg)	Regret (%)	Balanced (%)	Status
1	86.154	60.4161	-	valid
2	74.596	43.3580	-	valid
3	36.731	12.0802	-	valid
4	14.402	1.2506	-	valid
5	17.237	2.3451	-	valid
6	13.124	1.0264	-	valid
7	14.695	1.6725	-	valid
8	11.734	0.7778	-	valid
9	6.559	0.2156	-	valid
10	7.475	0.4226	-	valid
11	4.327	0.1078	-	valid
12	4.584	0.2240	-	valid
13	3.205	0.0392	-	valid
14	3.296	0.0774	-	valid
15	5.163	0.1621	-	valid
16	3.930	0.0804	-	valid
17	3.552	0.0887	-	valid
18	4.654	0.1125	-	valid
19	4.061	0.1078	-	valid
20	3.900	0.0582	-	valid

7. Stronger noise | seed 3 | Score base

t	Angle (deg)	Regret (%)	Balanced (%)	Status
1	52.600	9.5988	-	valid
2	61.776	36.5943	-	valid
3	41.015	20.4660	-	valid
4	18.512	1.9650	-	valid
5	11.464	1.4955	-	valid
6	9.027	0.4569	-	valid
7	12.885	1.3756	-	valid
8	5.637	0.2967	-	valid
9	11.283	1.1293	-	valid
10	7.210	0.4099	-	valid
11	9.736	0.4688	-	valid
12	5.726	0.2660	-	valid
13	7.504	0.3072	-	valid
14	7.316	0.3072	-	valid
15	6.082	0.1859	-	valid
16	5.011	0.1614	-	valid
17	6.789	0.1937	-	valid
18	9.067	0.3528	-	valid
19	6.943	0.1991	-	valid
20	9.766	0.3887	-	valid

Complete step records | runs 69-72 of 80

7. Stronger noise | seed 4 | Pedro

t	Angle (deg)	Regret (%)	Balanced (%)	Status
1	35.264	10.3284	-	valid
2	53.944	23.0108	-	valid
3	47.902	21.0013	-	valid
4	47.902	21.0013	-	valid
5	44.547	18.1474	-	valid
6	44.547	18.1474	-	valid
7	44.547	18.1474	-	valid
8	44.547	18.1474	-	valid
9	10.113	0.5725	-	valid
10	10.113	0.5725	-	valid
11	12.096	0.8497	-	valid
12	12.096	0.8497	-	valid
13	12.096	0.8497	-	valid
14	12.096	0.8497	-	valid
15	12.096	0.8497	-	valid
16	12.096	0.8497	-	valid
17	12.096	0.8497	-	valid
18	12.096	0.8497	-	valid
19	6.196	0.3991	-	valid
20	6.362	0.3126	-	valid

8. Query-dependent noise | seed 0 | Pedro

t	Angle (deg)	Regret (%)	Balanced (%)	Status
1	43.016	7.8862	-	valid
2	35.197	6.5878	-	valid
3	35.197	6.5878	-	valid
4	33.486	6.1289	-	valid
5	33.486	6.1289	-	valid
6	31.298	7.4432	-	valid
7	31.298	7.4432	-	valid
8	31.298	7.4432	-	valid
9	31.298	7.4432	-	valid
10	31.298	7.4432	-	valid
11	31.298	7.4432	-	valid
12	31.298	7.4432	-	valid
13	16.599	1.7693	-	valid
14	16.599	1.7693	-	valid
15	16.591	1.7693	-	valid
16	16.320	1.7693	-	valid
17	16.253	1.7815	-	valid
18	16.253	1.7815	-	valid
19	16.253	1.7815	-	valid
20	16.253	1.7815	-	valid

7. Stronger noise | seed 4 | Score base

t	Angle (deg)	Regret (%)	Balanced (%)	Status
1	37.997	12.1777	-	valid
2	35.344	7.9782	-	valid
3	7.657	0.2117	-	valid
4	5.719	0.3263	-	valid
5	8.108	0.7200	-	valid
6	7.457	0.5798	-	valid
7	6.825	0.3799	-	valid
8	10.744	1.2564	-	valid
9	8.634	0.7604	-	valid
10	9.100	0.5088	-	valid
11	4.757	0.3165	-	valid
12	3.788	0.2216	-	valid
13	2.395	0.0757	-	valid
14	0.845	0.0409	-	valid
15	2.385	0.1214	-	valid
16	1.792	0.0524	-	valid
17	1.719	0.0985	-	valid
18	2.820	0.1373	-	valid
19	2.500	0.1395	-	valid
20	2.309	0.1395	-	valid

8. Query-dependent noise | seed 0 | Score base

t	Angle (deg)	Regret (%)	Balanced (%)	Status
1	57.862	31.4900	-	valid
2	55.474	32.6131	-	valid
3	6.848	0.4335	-	valid
4	11.540	1.1461	-	valid
5	4.712	0.2858	-	valid
6	9.421	0.5682	-	valid
7	5.065	0.3181	-	valid
8	6.924	0.3809	-	valid
9	9.040	1.0881	-	valid
10	13.351	1.3416	-	valid
11	9.647	0.6190	-	valid
12	10.710	0.4812	-	valid
13	3.009	0.0380	-	valid
14	10.967	0.7991	-	valid
15	6.187	0.3083	-	valid
16	4.052	0.0892	-	valid
17	5.301	0.1046	-	valid
18	4.200	0.0926	-	valid
19	3.554	0.0926	-	valid
20	2.928	0.0926	-	valid

Complete step records | runs 73-76 of 80

8. Query-dependent noise | seed 1 | Pedro

t	Angle (deg)	Regret (%)	Balanced (%)	Status
1	76.919	59.2947	-	valid
2	22.519	2.0739	-	valid
3	34.130	11.9738	-	valid
4	34.130	11.9738	-	valid
5	41.220	19.7685	-	valid
6	38.449	15.7221	-	valid
7	38.449	15.7221	-	valid
8	38.505	15.9629	-	valid
9	14.951	1.3467	-	valid
10	14.951	1.3467	-	valid
11	14.951	1.3467	-	valid
12	14.951	1.3467	-	valid
13	14.951	1.3467	-	valid
14	14.951	1.3467	-	valid
15	14.951	1.3467	-	valid
16	20.837	2.5705	-	valid
17	20.837	2.5705	-	valid
18	20.837	2.5705	-	valid
19	20.837	2.5705	-	valid
20	20.837	2.5705	-	valid

8. Query-dependent noise | seed 2 | Pedro

t	Angle (deg)	Regret (%)	Balanced (%)	Status
1	60.453	25.8292	-	valid
2	33.129	4.3752	-	valid
3	36.685	7.1805	-	valid
4	32.927	7.7417	-	valid
5	33.373	7.7417	-	valid
6	33.373	7.7417	-	valid
7	21.282	2.9956	-	valid
8	21.282	2.9956	-	valid
9	21.282	2.9956	-	valid
10	21.282	2.9956	-	valid
11	21.282	2.9956	-	valid
12	21.282	2.9956	-	valid
13	21.282	2.9956	-	valid
14	21.282	2.9956	-	valid
15	21.282	2.9956	-	valid
16	21.282	2.9956	-	valid
17	21.282	2.9956	-	valid
18	21.282	2.9956	-	valid
19	21.282	2.9956	-	valid
20	21.282	2.9956	-	valid

8. Query-dependent noise | seed 1 | Score base

t	Angle (deg)	Regret (%)	Balanced (%)	Status
1	63.661	38.2588	-	valid
2	34.467	7.7085	-	valid
3	22.611	5.7869	-	valid
4	14.900	1.5448	-	valid
5	9.641	0.8047	-	valid
6	5.435	0.4191	-	valid
7	7.924	0.4155	-	valid
8	7.198	0.5009	-	valid
9	2.703	0.0558	-	valid
10	5.376	0.1767	-	valid
11	8.172	0.5734	-	valid
12	7.915	0.4458	-	valid
13	5.567	0.1792	-	valid
14	9.408	0.8534	-	valid
15	5.316	0.1254	-	valid
16	7.742	0.4014	-	valid
17	6.712	0.4428	-	valid
18	5.957	0.4284	-	valid
19	4.992	0.1668	-	valid
20	4.384	0.1254	-	valid

8. Query-dependent noise | seed 2 | Score base

t	Angle (deg)	Regret (%)	Balanced (%)	Status
1	86.154	60.4161	-	valid
2	74.596	43.3580	-	valid
3	36.731	12.0802	-	valid
4	14.402	1.2506	-	valid
5	17.237	2.3451	-	valid
6	13.124	1.0264	-	valid
7	14.695	1.6725	-	valid
8	11.734	0.7778	-	valid
9	6.559	0.2156	-	valid
10	7.475	0.4226	-	valid
11	4.327	0.1078	-	valid
12	4.584	0.2240	-	valid
13	3.205	0.0392	-	valid
14	3.296	0.0774	-	valid
15	5.163	0.1621	-	valid
16	3.930	0.0804	-	valid
17	3.552	0.0887	-	valid
18	4.654	0.1125	-	valid
19	4.061	0.1078	-	valid
20	3.900	0.0582	-	valid

Complete step records | runs 77-80 of 80

8. Query-dependent noise | seed 3 | Pedro

t	Angle (deg)	Regret (%)	Balanced (%)	Status
1	46.067	9.1259	-	valid
2	29.993	3.2335	-	valid
3	11.936	1.2699	-	valid
4	26.368	3.9823	-	valid
5	30.006	6.1538	-	valid
6	30.006	6.1538	-	valid
7	18.575	3.0839	-	valid
8	10.450	0.5109	-	valid
9	10.450	0.5109	-	valid
10	11.540	0.7064	-	valid
11	11.540	0.7064	-	valid
12	11.540	0.7064	-	valid
13	11.540	0.7064	-	valid
14	11.540	0.7064	-	valid
15	11.540	0.7064	-	valid
16	11.540	0.7064	-	valid
17	11.532	1.0970	-	valid
18	11.865	1.1891	-	valid
19	11.865	1.1891	-	valid
20	11.865	1.1891	-	valid

8. Query-dependent noise | seed 4 | Pedro

t	Angle (deg)	Regret (%)	Balanced (%)	Status
1	35.264	10.3284	-	valid
2	53.944	23.0108	-	valid
3	47.902	21.0013	-	valid
4	47.902	21.0013	-	valid
5	44.547	18.1474	-	valid
6	44.547	18.1474	-	valid
7	44.547	18.1474	-	valid
8	44.547	18.1474	-	valid
9	10.113	0.5725	-	valid
10	10.113	0.5725	-	valid
11	12.096	0.8497	-	valid
12	12.096	0.8497	-	valid
13	12.096	0.8497	-	valid
14	12.096	0.8497	-	valid
15	12.096	0.8497	-	valid
16	12.096	0.8497	-	valid
17	12.096	0.8497	-	valid
18	12.096	0.8497	-	valid
19	6.196	0.3991	-	valid
20	6.362	0.3126	-	valid

8. Query-dependent noise | seed 3 | Score base

t	Angle (deg)	Regret (%)	Balanced (%)	Status
1	52.600	9.5988	-	valid
2	61.776	36.5943	-	valid
3	41.015	20.4660	-	valid
4	18.512	1.9650	-	valid
5	11.464	1.4955	-	valid
6	9.027	0.4569	-	valid
7	12.885	1.3756	-	valid
8	5.637	0.2967	-	valid
9	11.283	1.1293	-	valid
10	7.210	0.4099	-	valid
11	9.736	0.4688	-	valid
12	5.726	0.2660	-	valid
13	7.504	0.3072	-	valid
14	7.316	0.3072	-	valid
15	6.082	0.1859	-	valid
16	6.001	0.2811	-	valid
17	4.041	0.1598	-	valid
18	3.083	0.0256	-	valid
19	4.873	0.1114	-	valid
20	4.709	0.1114	-	valid

8. Query-dependent noise | seed 4 | Score base

t	Angle (deg)	Regret (%)	Balanced (%)	Status
1	37.997	12.1777	-	valid
2	35.344	7.9782	-	valid
3	7.657	0.2117	-	valid
4	5.719	0.3263	-	valid
5	8.108	0.7200	-	valid
6	7.457	0.5798	-	valid
7	6.825	0.3799	-	valid
8	10.744	1.2564	-	valid
9	8.634	0.7604	-	valid
10	4.443	0.1679	-	valid
11	4.088	0.1580	-	valid
12	3.186	0.0758	-	valid
13	5.005	0.3165	-	valid
14	3.159	0.1381	-	valid
15	4.780	0.1931	-	valid
16	3.852	0.1586	-	valid
17	4.377	0.2278	-	valid
18	2.125	0.0758	-	valid
19	2.953	0.2018	-	valid
20	3.747	0.0963	-	valid